Practical: -0

Q] Introduction to Android, Introduction to Android Studio IDE, Application Fundamentals: Creating a Project, Android Components, Activities, Services, Content Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB debugging mode, Android Application Overview. Simple “Hello World” program.

Introduction to Android

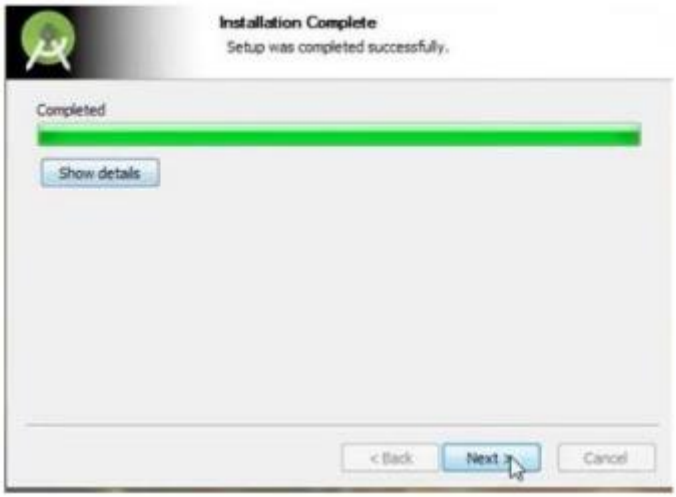
Android is an open-source mobile operating system developed by Google, primarily used for smartphones and tablets. It is built on the Linux kernel and allows developers to create apps using languages like Java and Kotlin. Android supports a wide variety of applications, ranging from simple utilities to complex games, and offers a flexible platform for developers.

Introduction to Android Studio IDE

Android Studio is the official IDE for Android app development. Developed by Google, it provides all the tools needed to build, test, and debug Android applications. Android Studio supports both Java and Kotlin, includes a powerful code editor, an emulator for testing apps, a UI designer, and various performance tools to streamline the development process.

Installation

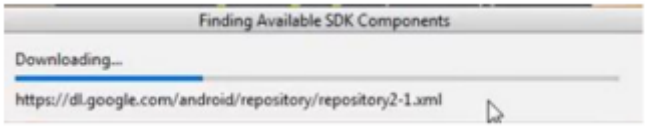
Welcome to Android Studio Setup
Setup will guide you through the installation of Android Studio.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Android Studio
< Back
Next >
Cancel


Installation Complete
Setup was completed successfully.
Completed
Show details
< Back
Next >
Cancel

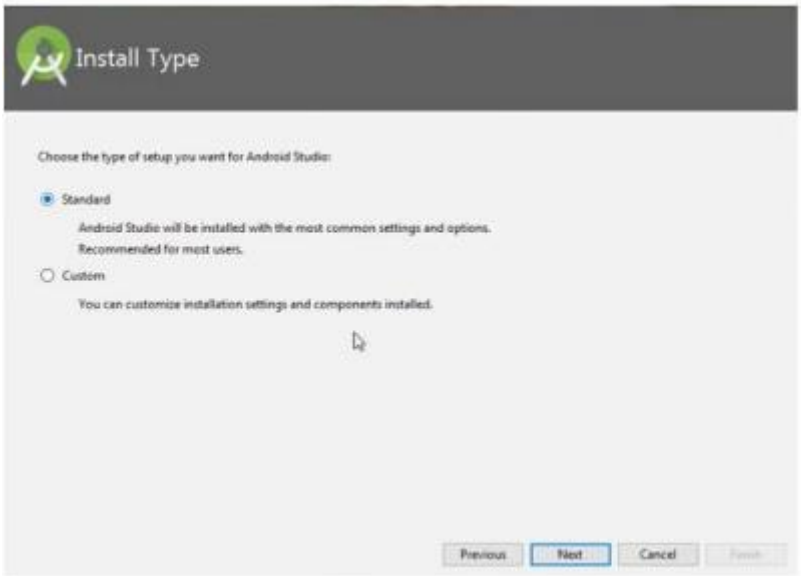
Completing Android Studio Setup
Android Studio has been installed on your computer.
Click Finish to close Setup.
Start Android Studio
Android Studio
< Back
Finish
Cancel

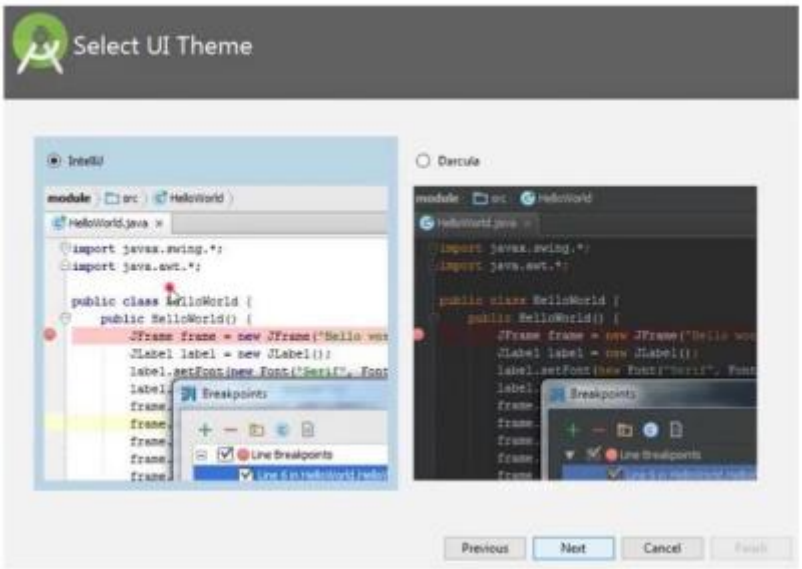
Import Studio settings from:
Custom location, Config folder or installation home of the previous version:
Do not import settings
OK

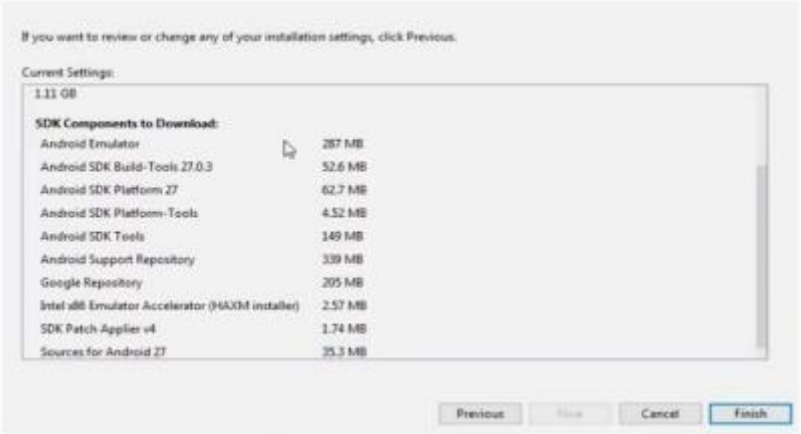
android studio
Powered by the IntelliJ® Platform

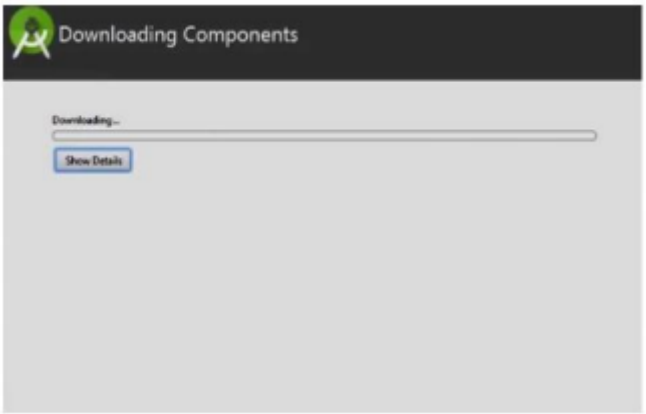

Finding Available SDK Components
Downloading...
https://dl.google.com/android/repository/repository2-1.xml

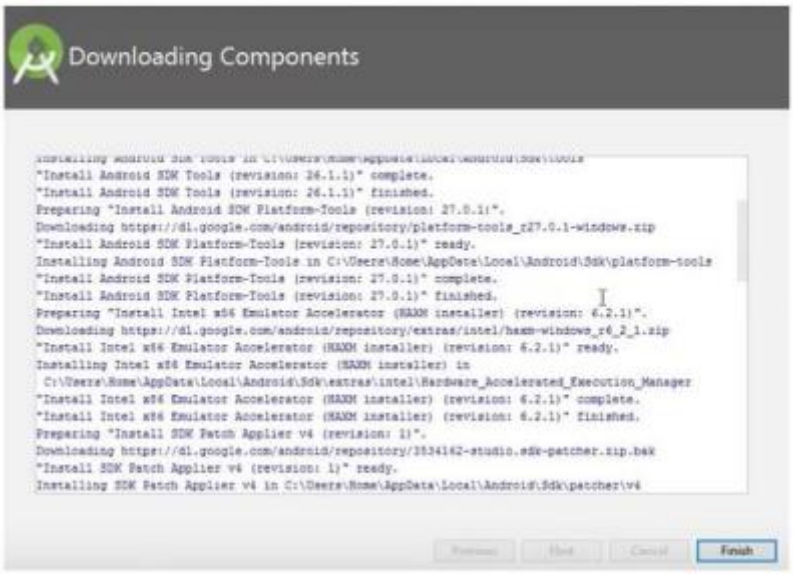
Welcome
Android Studio
Welcome! This wizard will set up your development environment for Android Studio.
Additionally, the wizard will help port existing Android apps into Android Studio
or create a new Android application project.
Next
Cancel


Install Type
Choose the type of setup you want for Android Studio:
Standard
Android Studio will be installed with the most common settings and options.
Recommended for most users.
Custom
You can customize installation settings and components installed.
Previous
Next
Cancel

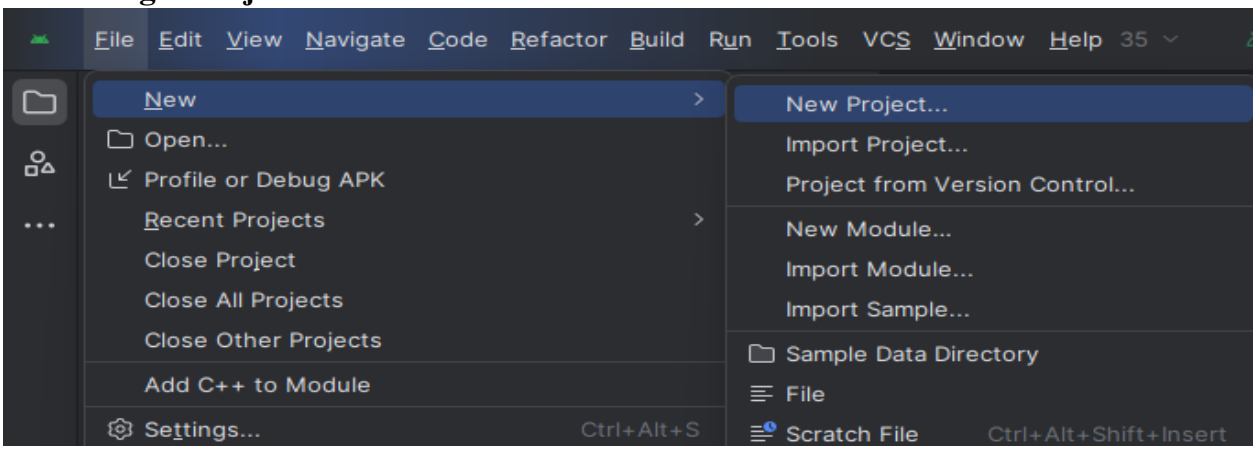

Select UI Theme
IntelliJ
Darcula
Previous
Next
Cancel

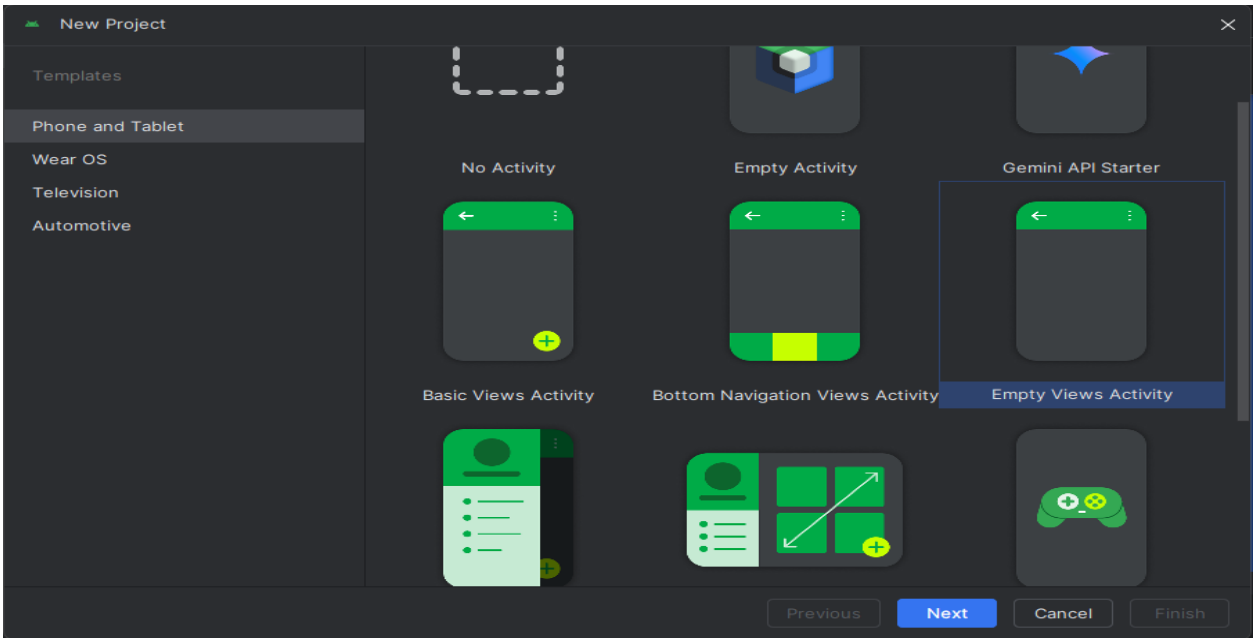

If you want to review or change any of your installation settings, click Previous.
Current Settings:
1.11 GB
SDK Components to Download:
Android Emulator 287 MB
Android SDK Build-Tools 27.0.3 52.6 MB
Android SDK Platform 27 62.7 MB
Android SDK Platform-Tools 4.52 MB
Android SDK Tools 149 MB
Android Support Repository 339 MB
Google Repository 205 MB
Intel x86 Emulator Accelerator (HAXM installer) 2.57 MB
SDK Patch Applier v4 1.74 MB
Sources for Android 27 35.3 MB
Previous
Cancel
Finish

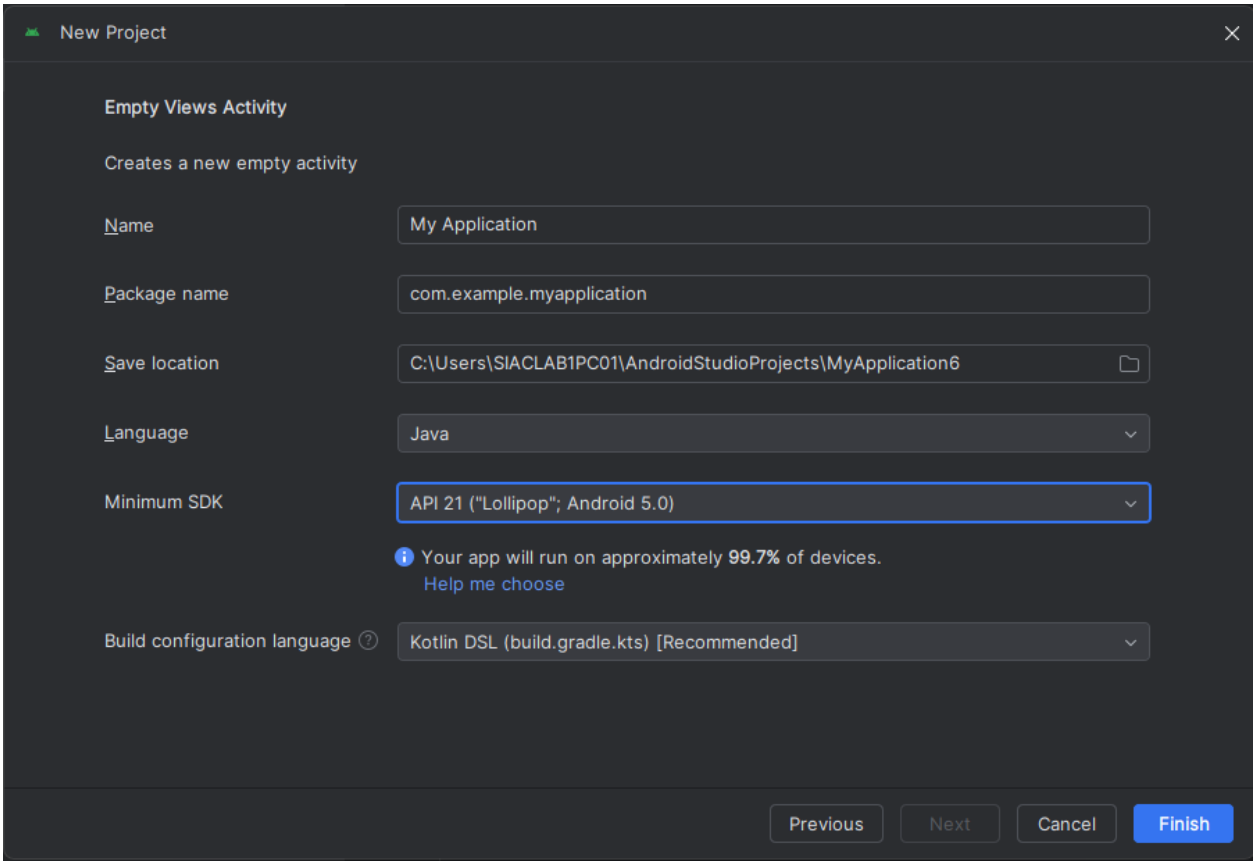

Downloading Components
Downloading...
Show Details



a. Creating a Project

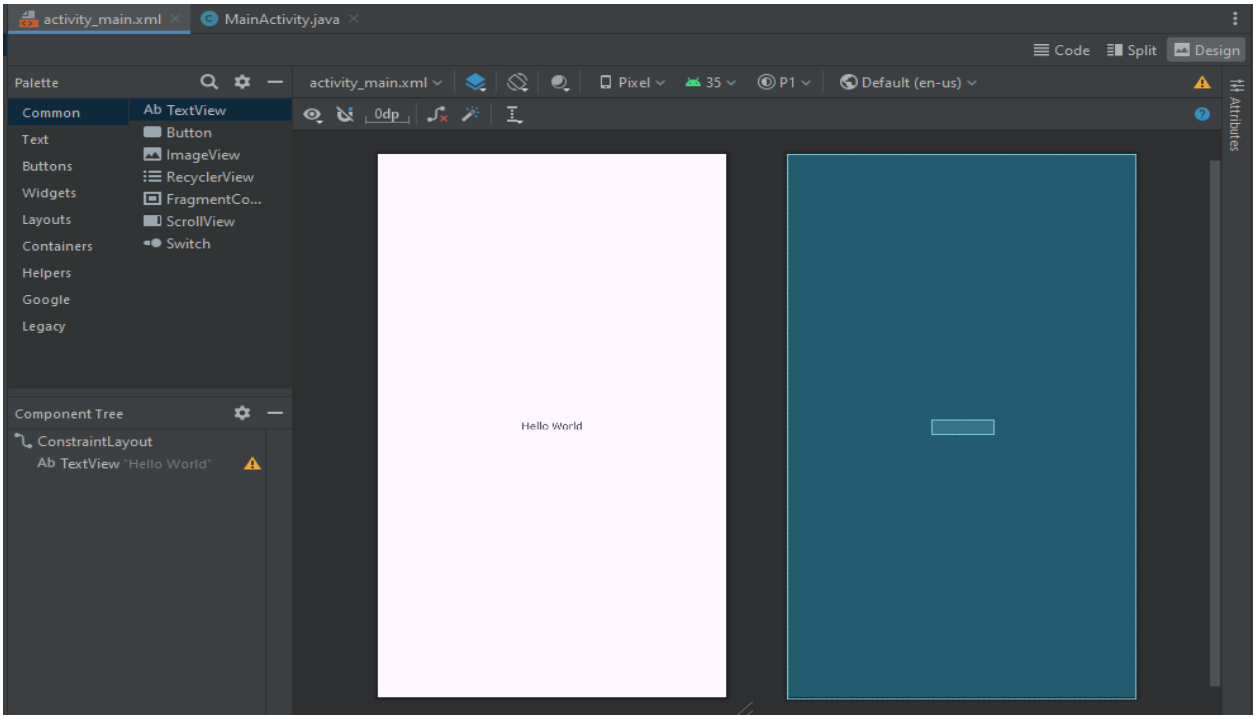






b. Simple "Hello World" program (activity_main.xml)

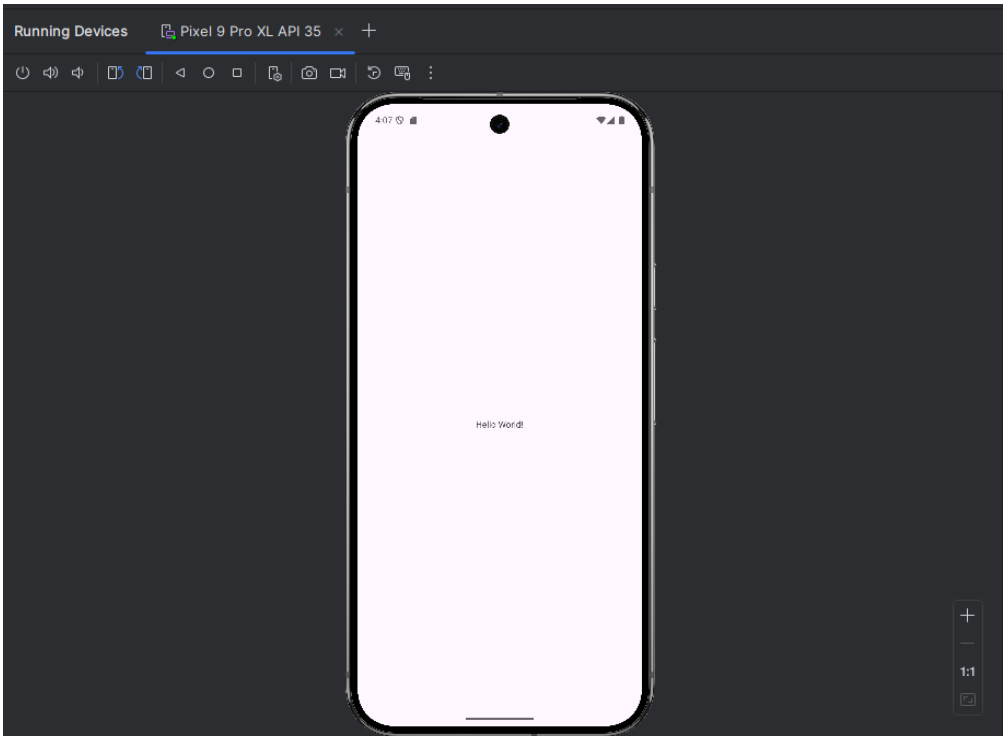
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```



(MainActivity.java)

```
package com.example.p1;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

OUTPUT



Practical: -1

Q] **Programming Resources Android Resources:**
(Color, Theme, String, Drawable, Dimension, Image)

a. **Defining Color Property**

Create a new project and go to: ProjectName>App>src>main>res>values>colors.xml (Colours.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <color name="black">#FF000000</color>
   <color name="white">#FFFFFFFF</color>
   <color name="red">#FF0000</color>
   <color name="Red">#FF0000</color>
   <color name="Blue">#0000FF</color>
   <color name="Pink">#FF00FF</color>
   <color name="Cyan">#399036</color>
   <color name="Grey">#AABBFF</color>
   <color name="colorPrimary">#008577</color>
   <color name="colorPrimaryDark">#00574B</color>
   <color name="colorAccent">#D81B60</color>
</resources>
```

b. **Defining Theme Property**

Create a new project and go to: ProjectName>App>src>main>res>values>styles.xml (Styles.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
      <!-- Customize your theme here. -->
      <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      <item name="android:background">#000000</item>
      <item name="android:textColor">#FFFFAA</item>
   </style>
</resources>
```

c. **Defining string property**

Create a new project and go to: ProjectName>App>src>main>res>values>strings.xml (Strings.xml)

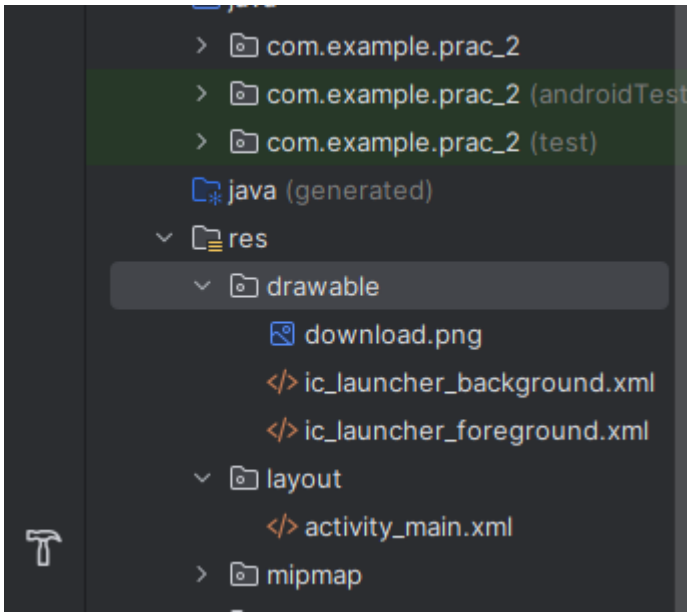
```
<resources>
   <string name="app_name">PRAC_2</string>
   <string name="Heading">Programming Resources</string>
   <string name="Description">
 Android is a mobile operating system developed by Google.
 It is based on a modified version of the Linux
 kernel and other open source software, and is
```

```
 designed primarily for touch screen mobile devices
 such as smart phones and tablets
</string>
</resources>
```

d. **Adding images and dimensions**

For adding a new image file, do the following:

ProjectName>App>src>main>res>drawable>Right-click and paste the images



Dim.xml

```
<?xml version="1.0" encoding="utf-8"?>
  <resources>
<dimen name="textview_height">35dp</dimen>
<dimen name="textview_width">150dp</dimen>
<dimen name="font_size">26sp</dimen>
  </resources>
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello World!"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
```

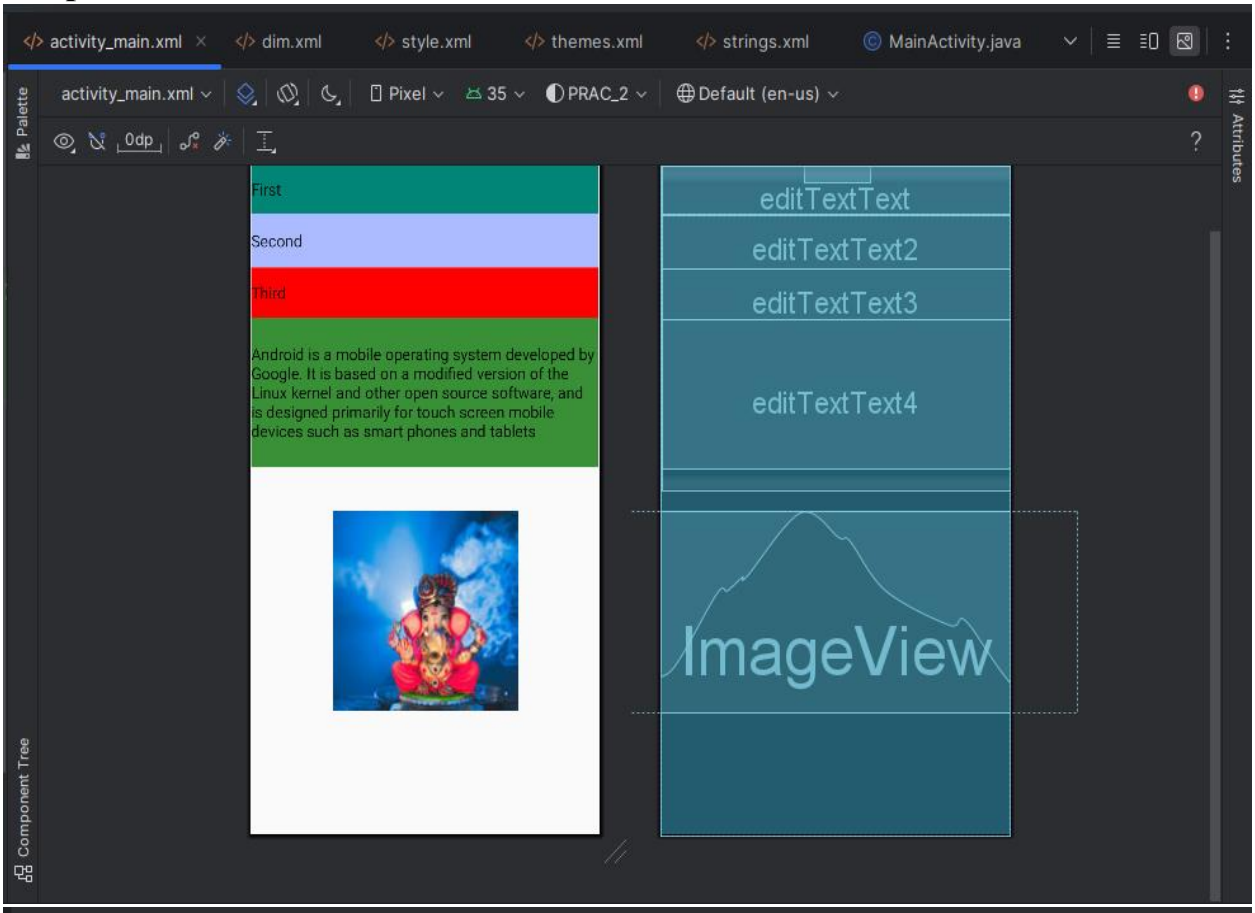
```
    app:layout_constraintTop_toTopOf="parent" />
<LinearLayout
    android:id="@+id/linearLayout"
    android:layout_width="409dp"
    android:layout_height="354dp"
    android:orientation="vertical"
    tools:layout_editor_absoluteX="1dp"
    tools:layout_editor_absoluteY="1dp">
    <EditText
        android:id="@+id/editTextText"
        android:layout_width="match_parent"
        android:layout_height="53dp"
        android:background="@color/colorPrimary"
        android:ems="10"
        android:inputType="text"
        android:text="First" />
    <EditText
        android:id="@+id/editTextText2"
        android:layout_width="match_parent"
        android:layout_height="59dp"
        android:background="@color/Grey"
        android:ems="10"
        android:inputType="text"
        android:text="Second" />
    <EditText
        android:id="@+id/editTextText3"
        android:layout_width="match_parent"
        android:layout_height="55dp"
        android:background="@color/red"
        android:ems="10"
        android:inputType="text"
        android:text="Third" />
    <EditText
        android:id="@+id/editTextText4"
        android:layout_width="match_parent"
        android:layout_height="163dp"
        android:background="@color/Cyan"
        android:ems="10"
        android:text="@string/Description" />
</LinearLayout>
<ImageView
    android:id="@+id/imageView"
    android:layout_width="565dp"
    android:layout_height="219dp"
    android:layout_marginTop="35dp"
    android:layout_marginEnd="249dp"
    android:layout_marginBottom="193dp"
```

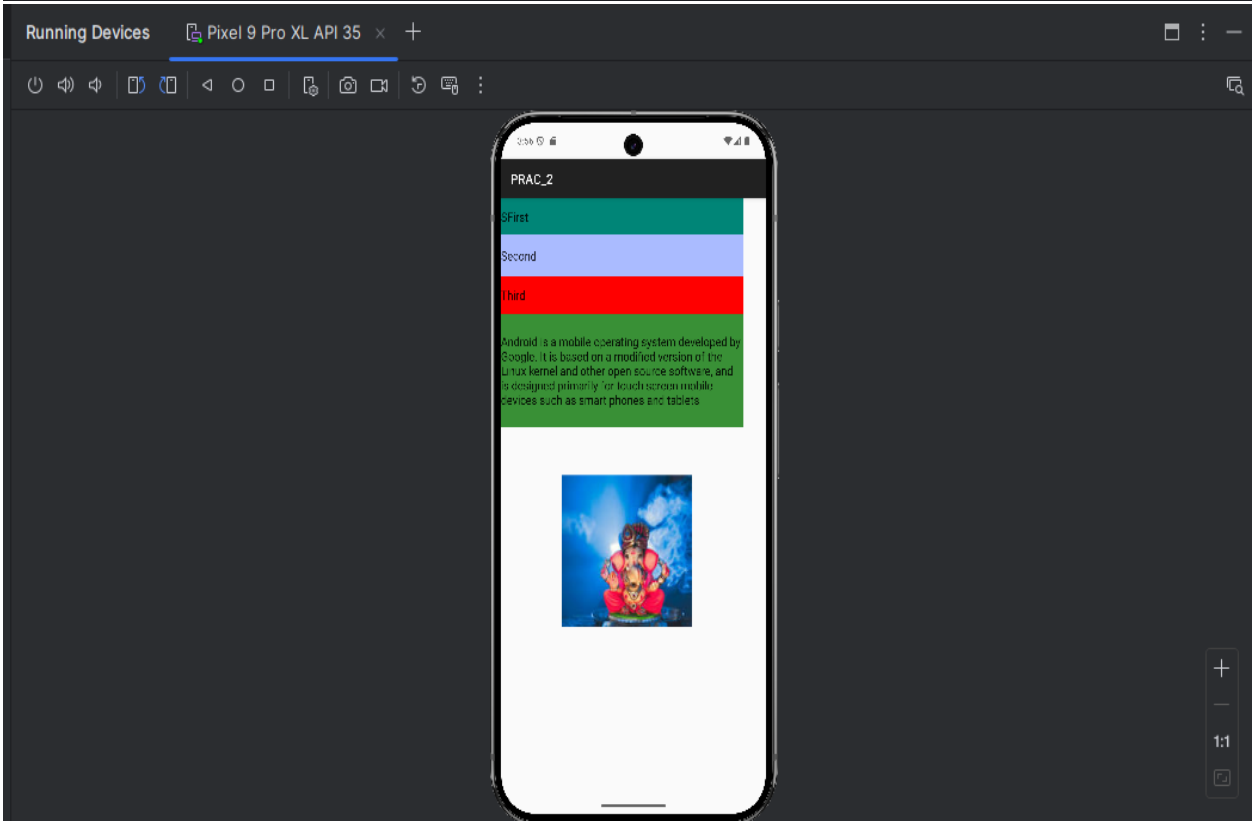
```
        android:visibility="visible"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.191"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/linearLayout"
        app:layout_constraintVertical_bias="0.169"
        app:srcCompat="@drawable/download" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

(Mainactivity.java)

```
package com.example.prac_2;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

Output:

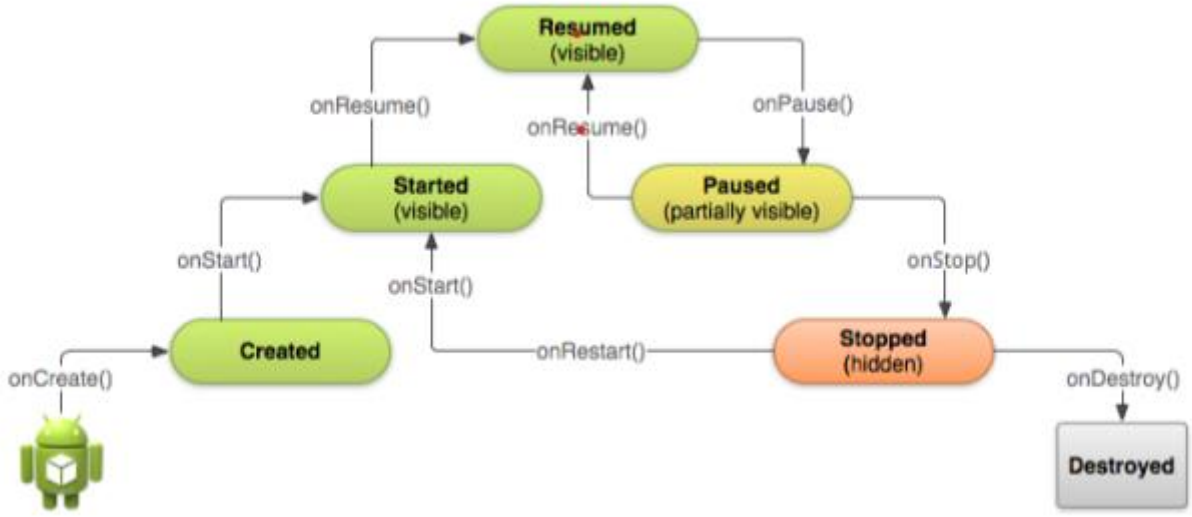

activity_main.xml
dim.xml
style.xml
themes.xml
strings.xml
MainActivity.java
Pixel
35
PRAC_2
Default (en-us)
Palette
Attributes
Component Tree
First
Second
Third
Android is a mobile operating system developed by Google. It is based on a modified version of the Linux kernel and other open source software, and is designed primarily for touch screen mobile devices such as smart phones and tablets
editTextText
editTextText2
editTextText3
editTextText4
ImageView


Running Devices
Pixel 9 Pro XL API 35
PRAC_2
First
Second
Third
1:1

Practical: -2

Q] Programming Activities and fragments

Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.



1. onCreate(): Called by the OS when the activity is first created. This is where you initialize any UI elements or data objects. You also have the saved InstanceState of the activity that contains its previously saved state, and you can use it to recreate that state.
2. onStart(): Just before presenting the user with an activity, this method is called. It's always followed by onResume(). In here, you generally should start UI animations, audio based content or anything else that requires the activity's contents to be on screen.
3. onResume(): As an activity enters the foreground, this method is called. Here you have a good place to restart animations, update UI elements, restart camera previews, resume audio/video playback or initialize any components that you release during onPause()
4. onPause(): This method is called before sliding into the background. Here you should stop any visuals or audio associated with the activity such as UI animations, music playback or the camera. This method is followed by onResume() if the activity returns to the foreground or by onStop() if it becomes hidden
5. onStop(): This method is called right after onPause(), when the activity is no longer visible to the user, and it's a good place to save data that you want to commit to the disk. It's followed by either onRestart(), if this activity is coming back to the foreground, or onDestroy() if it's being released from memory.
6. onRestart(): Called after stopping an activity, but just before starting it again. It's always followed by onStart().
7. onDestroy(): This is the final callback you'll receive from the OS before the activity is destroyed. You can trigger an activity's destruction by calling finish(), or it can be triggered by the system when the system needs to recoup memory. If your activity includes any background threads or other long-running resources, destruction could lead to a memory leak if they're not released, so you need to remember to stop these processes here as well.

Lifecycle
(MainActivity.java)

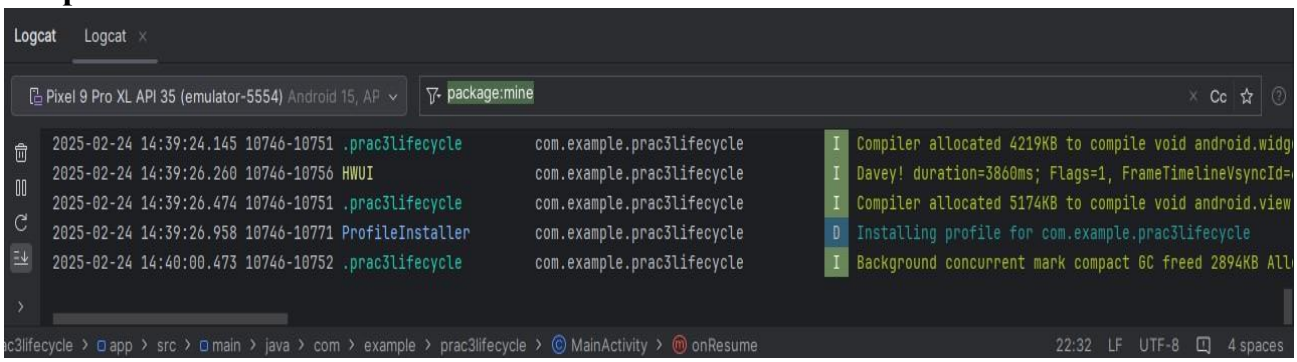
```
package com.example.prac3lifecycle; import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends
AppCompatActivity { private static final String TAG = "ActivityLifecycle";
@Override
protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
// Log message when the activity is created Log.d(TAG, "onCreate() called");
}
@Override
protected void onStart() { super.onStart();
// Log message when the activity becomes visible Log.d(TAG, "onStart() called");
}
@Override
protected void onResume() {
super.onResume();
// Log message when the activity is resumed and in the foreground
 Log.d(TAG, "onResume() called");
}
@Override
protected void onPause() { super.onPause();
// Log message when the activity is paused Log.d(TAG, "onPause() called");
}
@Override
protected void onStop() { super.onStop();
// Log message when the activity is no longer visible Log.d(TAG, "onStop() called");
}
@Override
protected void onDestroy() { super.onDestroy();
// Log message when the activity is being destroyed
Log.d(TAG, "onDestroy() called");
}
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".MainActivity">
<TextView
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:text="Hello World!"
```

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent" app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:-



Life Cycle of fragments and multiple fragments

1.Create a new project
2.Add a new Fragment ProjectName>App>src>main>Java
3. Right click on Java and select 'Add Fragment'>Add Fragment(Blank)
4.Give a name to the fragment and click Finish
5.Go to the new file and press Alt + Insert and click on 'Generate'>Override Methods **(**Or directly press Ctrl+O**)**
6.Select the lifecycle methods that we need to override, which are as follows:

- onAttach(context:Context):void
- onCreate(savedInstanceState:Bundle):void
- onViewCreated(view:View savedInstanceState:Bundle):void
- onActivityCreated(savedInstanceState:Bundle):void
- onStart():void
- onResume():void
- onPause():void
- onStop():void
- onDestroyView():void
- onDestroy():void

7.Now go to, LifeOfFragment.java(New Fragment File)

(MainActivity.java)

```
package com.example.prac3fragment;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentTransaction;
public class MainActivity extends AppCompatActivity {
   @Override
    protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
    // Check if the fragment is already added to avoid adding it again if re-created
   if  (savedInstanceState == null) {
   // Start a new fragment transaction to add MyFragment
    }
  }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent">
<!-- Fragment container -->
<FrameLayout
android:id="@+id/fragment_container"
android:layout_width="match_parent"
android:layout_height="match_parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

(MyFragment.java)

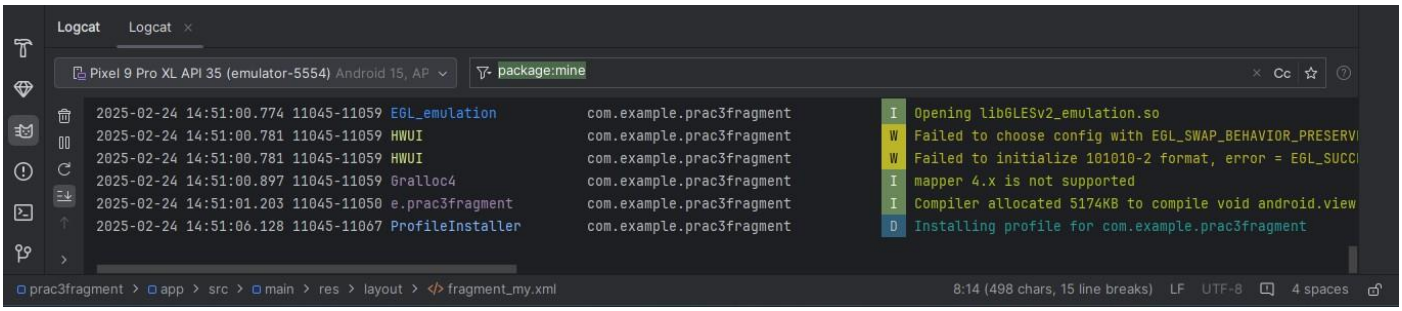
```
package com.example.prac3fragment; import android.os.Bundle;
import androidx.fragment.app.Fragment; import android.view.LayoutInflater; import
android.view.View;
import android.view.ViewGroup;
/**
* A simple {@link Fragment} subclass.
* Use the {@link MyFragment#newInstance} factory method to
* create an instance of this fragment.
*/
import android.os.Bundle;
import android.view.LayoutInflater; import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment; public class MyFragment extends Fragment {
@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_my, container, false);
}
}
```

(fragment_my.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent" android:orientation="vertical"
```

```
android:gravity="center">
<TextView
android:id="@+id/fragment_text" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Hello from MyFragment"
android:textSize="18sp"/>
</LinearLayout>
```

Output:-



Practical: -3

Q] Programs related to different Layouts

Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

Layouts in Android Studio control the positioning and arrangement of UI elements on the screen.

- Coordinate Layout: Flexible and dynamic, often used with advanced UI behaviors.
- Linear Layout**:** Simple, linear arrangement (either vertical or horizontal).
- Relative Layout: Positions views relative to others or the parent container.
- Table Layout**:** Organizes views in rows and columns, like a table.
- Absolute Layout (deprecated): Positions views at fixed pixel locations (no longer recommended).
- Frame Layout**:** Stacks views on top of each other.
- List View: Displays scrollable lists of data.
- Grid View**:** Displays items in a grid format.

a.Linear

(MainActivity.java)

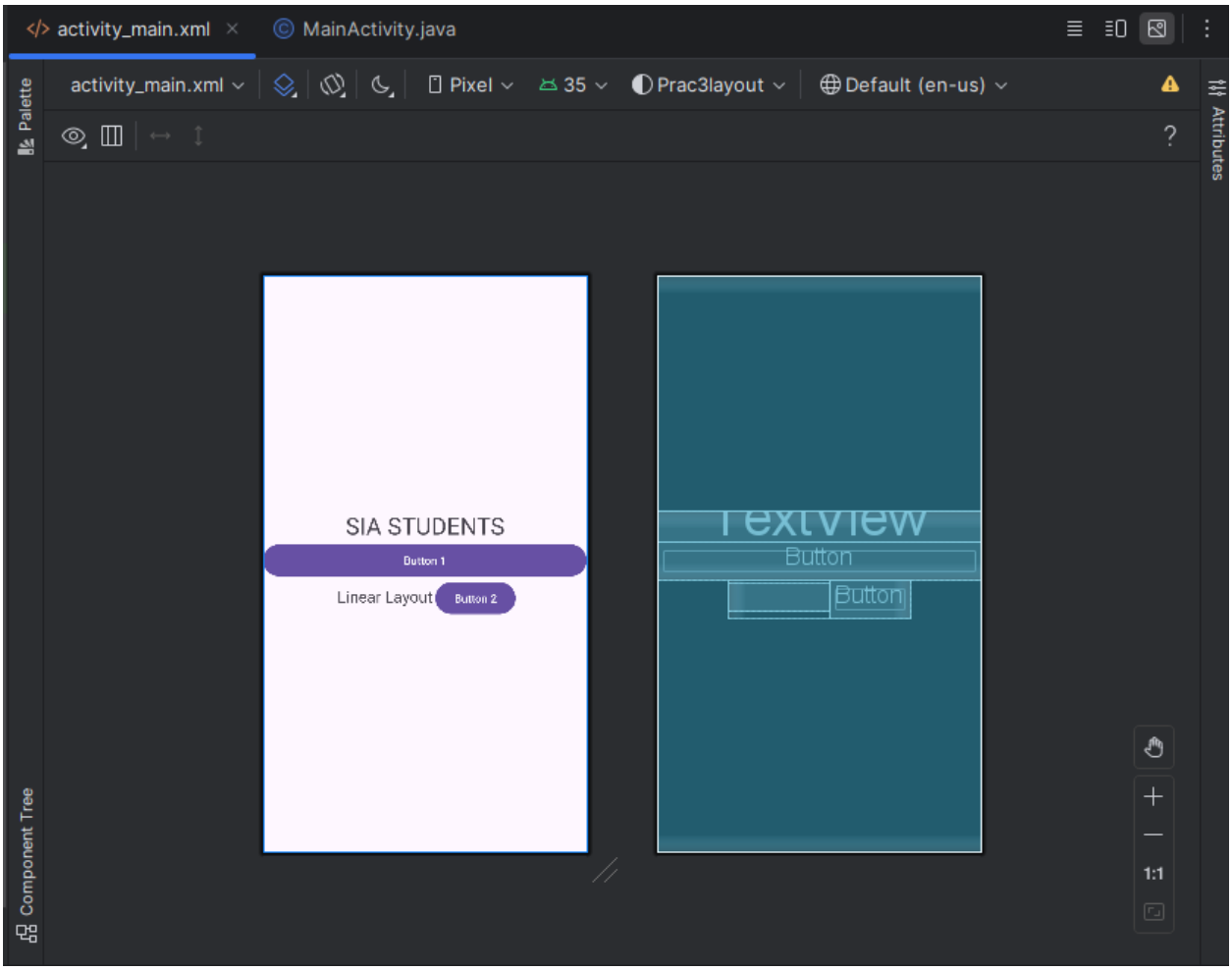
```
package com.example.prac3layout;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
   private Button btnClick;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      btnClick = (Button) findViewById(R.id.btn1);
      btnClick.setOnClickListener(new View.OnClickListener() {
         public void onClick(View view)
         {
            Toast.makeText(getApplicationContext(), "Button 1 CLicked",
Toast.LENGTH_SHORT).show();
         }
```

```
        });
        btnClick = (Button) findViewById(R.id.btn2);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 2 CLicked",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        >
        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center"
            android:textAlignment="center"
            android:textSize="30dp"
            android:text="SIA STUDENTS" />
        <Button
            android:id="@+id/btn1"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Button 1" />
    </LinearLayout>
    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        >
        <TextView
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textAlignment="center"
        android:padding="10px"
        android:textSize="20dp"
        android:text="Linear Layout" />
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2" />
  </LinearLayout>
</LinearLayout>
```



b.Relative

(MainActivity.java)

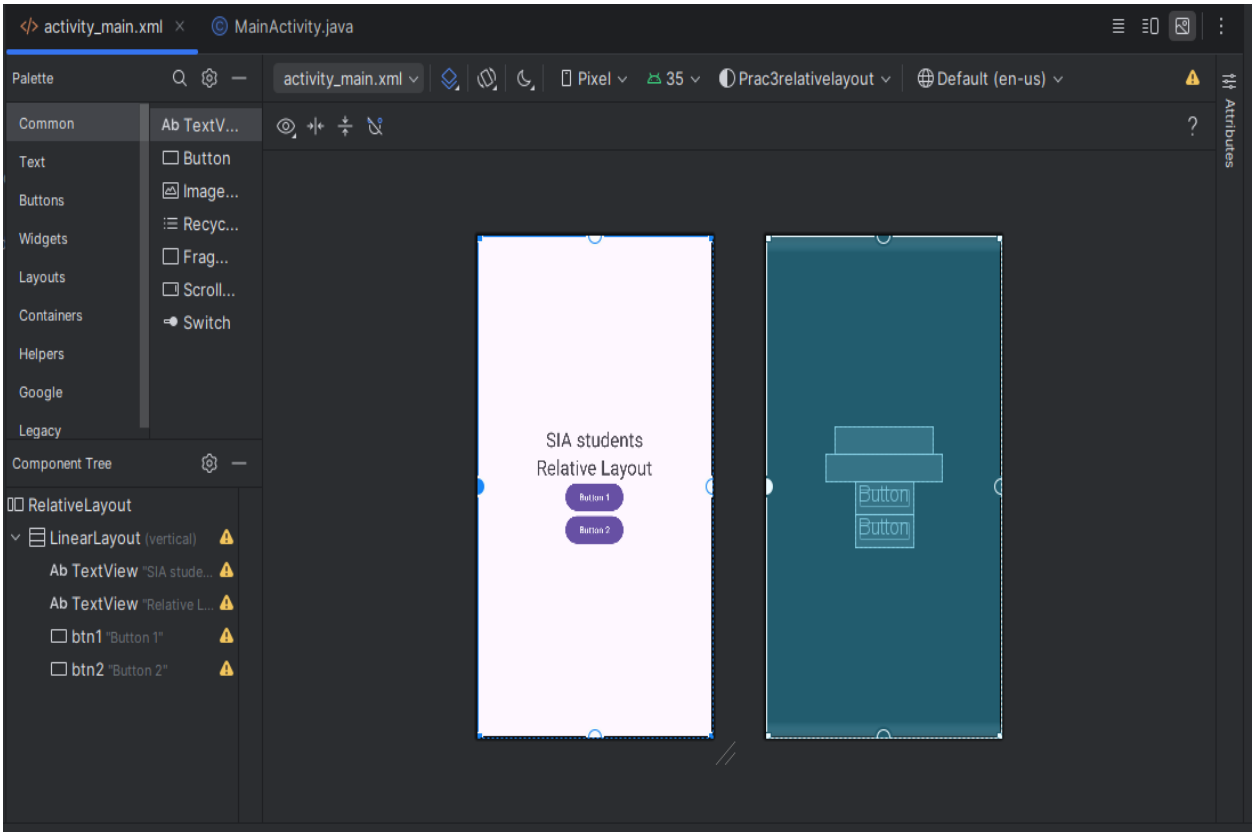
```
package com.example.prac3relativelayout;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

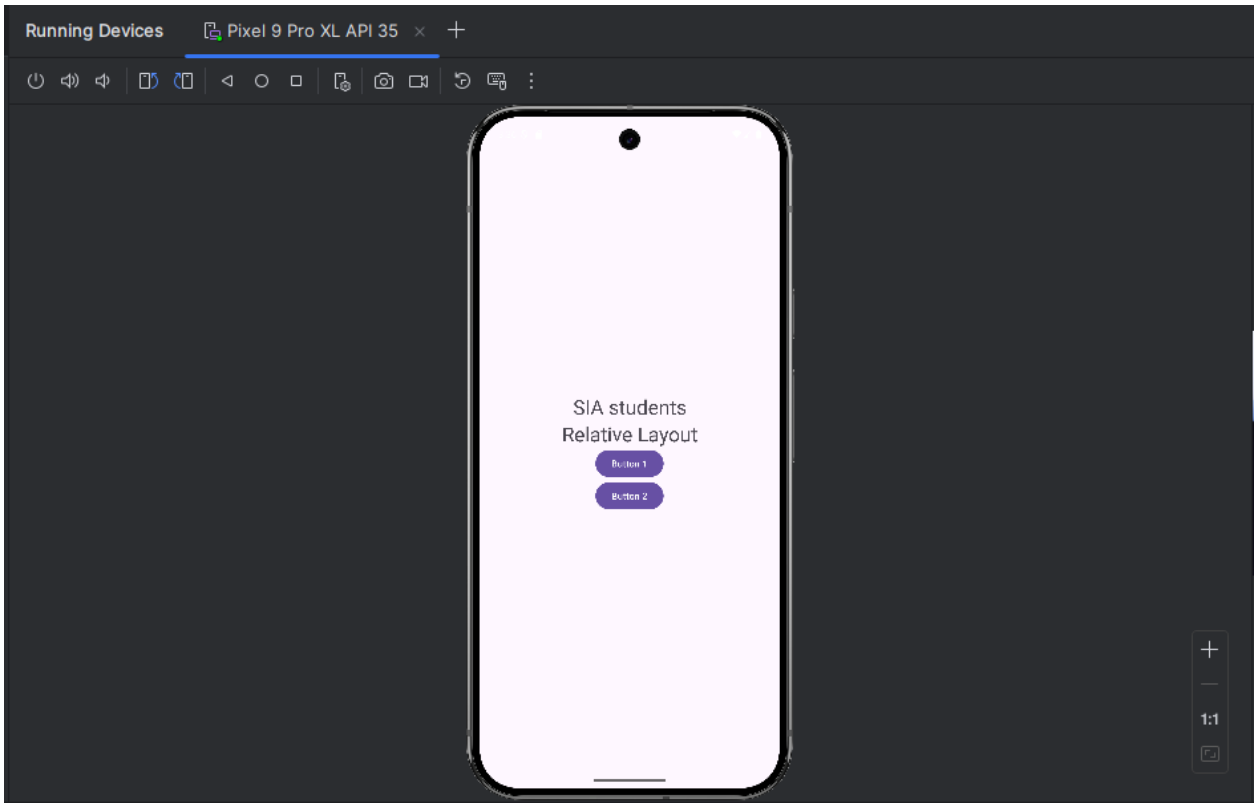
```
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast
public class MainActivity extends AppCompatActivity {
    private Button btnClick;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnClick = (Button) findViewById(R.id.btn1);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 1 Clicked",
Toast.LENGTH_SHORT).show();
            }
        });
        btnClick = (Button) findViewById(R.id.btn2);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "Button 2 Clicked",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    >
    <LinearLayout
        android:orientation="vertical"
        android:layout_width="fill_parent"
        android:layout_height="fill_parent"
        android:layout_alignParentStart="true"
        android:gravity="center">
```

```
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textAlignment="center"
            android:textSize="30dp"
            android:text="SIA students" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textAlignment="center"
            android:textSize="30dp"
            android:text="Relative Layout" />
        <Button
            android:id="@+id/btn1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 1" />
        <Button
            android:id="@+id/btn2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 2" />
    </LinearLayout>
</RelativeLayout>
```





c. Table

(MainActivity.java)

```
package com.example.tablelayout4c;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    private Button btnClick;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnClick = (Button) findViewById(R.id.btn1);
        btnClick.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View view)
            {
                Toast.makeText(getApplicationContext(), "1", Toast.LENGTH_SHORT).show();
            }
```

```
});
btnClick = (Button) findViewById(R.id.btn2);
btnClick.setOnClickListener(new View.OnClickListener()
{
   public void onClick(View view)
   {
      Toast.makeText(getApplicationContext(), "2", Toast.LENGTH_SHORT).show();
   }
});
btnClick = (Button) findViewById(R.id.btn3);
btnClick.setOnClickListener(new View.OnClickListener()
{
   public void onClick(View view)
   {
      Toast.makeText(getApplicationContext(), "3", Toast.LENGTH_SHORT).show();
   }
});
btnClick = (Button) findViewById(R.id.btn4);
btnClick.setOnClickListener(new View.OnClickListener()
{
   public void onClick(View view) {
      Toast.makeText(getApplicationContext(), "4", Toast.LENGTH_SHORT).show();
   }
});
      btnClick = (Button) findViewById(R.id.btn5);
      btnClick.setOnClickListener(new View.OnClickListener()
      {
         public void onClick(View view)
         {
            Toast.makeText(getApplicationContext(), "5", Toast.LENGTH_SHORT).show();
         }
      });
      btnClick = (Button) findViewById(R.id.btn6);
      btnClick.setOnClickListener(new View.OnClickListener()
      {
         public void onClick(View view)
         {
            Toast.makeText(getApplicationContext(), "6", Toast.LENGTH_SHORT).show();
         }
      });
      btnClick = (Button) findViewById(R.id.btn7);
      btnClick.setOnClickListener(new View.OnClickListener()
      {
         public void onClick(View view)
         {
            Toast.makeText(getApplicationContext(), "7", Toast.LENGTH_SHORT).show();
         }
```

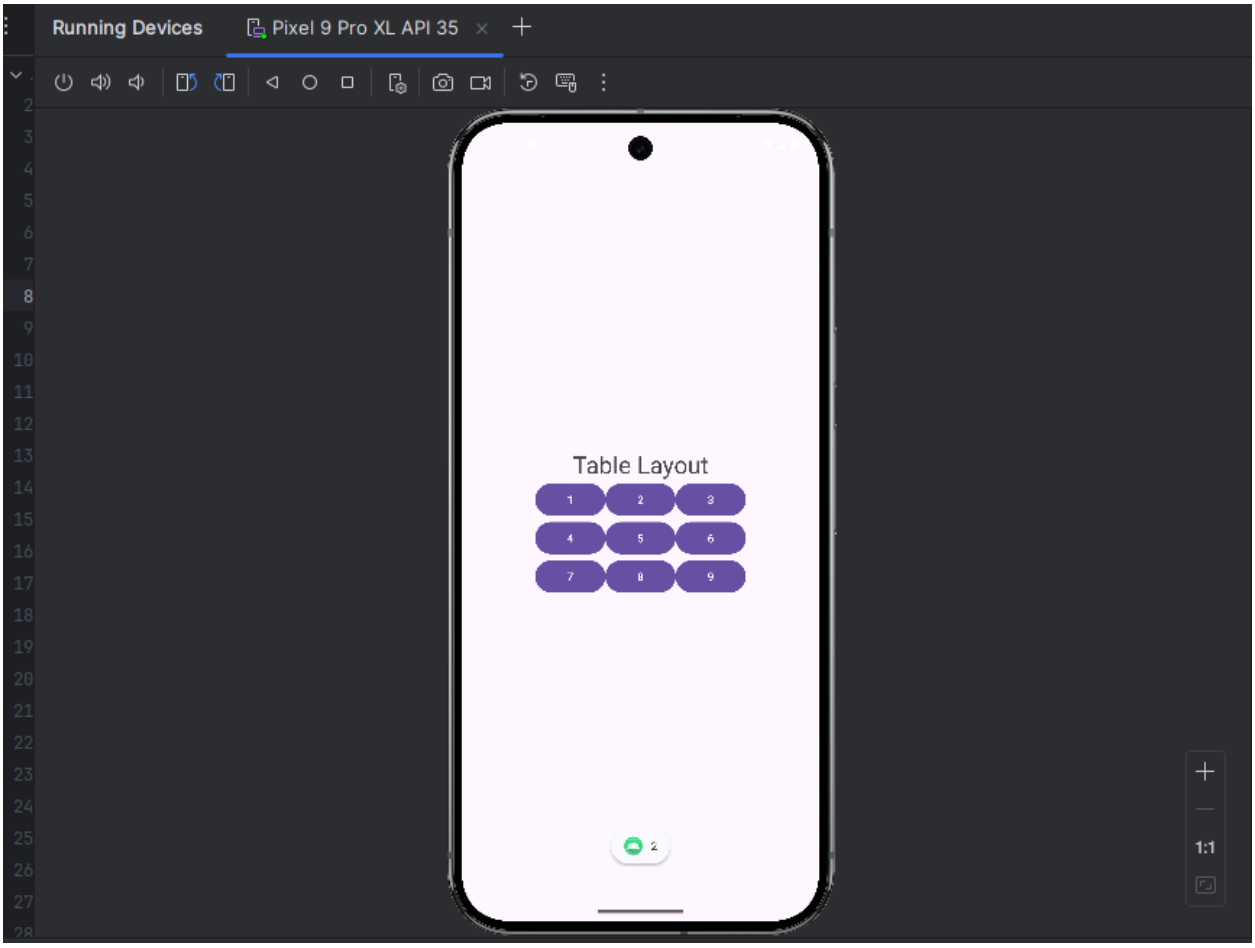
```
            });
            btnClick = (Button) findViewById(R.id.btn8);
            btnClick.setOnClickListener(new View.OnClickListener()
            {
                public void onClick(View view)
                {
                    Toast.makeText(getApplicationContext(), "8", Toast.LENGTH_SHORT).show();
                }
            });
            btnClick = (Button) findViewById(R.id.btn9);
            btnClick.setOnClickListener(new View.OnClickListener()
            {
                public void onClick(View view)
                {
                    Toast.makeText(getApplicationContext(), "9", Toast.LENGTH_SHORT).show();
                }
            });
        }
    }
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="30dp"
        android:text="Table Layout" />
    <TableLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">
        <TableRow
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">
            <Button
                android:id="@+id/btn1"
                android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="1" />
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="2" />
    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="3" />
</TableRow>
<TableRow
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">
    <Button
        android:id="@+id/btn4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="4" />
    <Button
        android:id="@+id/btn5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="5" />
    <Button
        android:id="@+id/btn6"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="6" />
</TableRow>
<TableRow
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">
    <Button
        android:id="@+id/btn7"
        android:layout_width="wrap_content
        android:layout_height="wrap_content"
        android:text="7" />
    <Button
        android:id="@+id/btn8"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="8" />
    <Button
        android:id="@+id/btn9"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="9" />
        </TableRow>
    </TableLayout>
</LinearLayout>
```



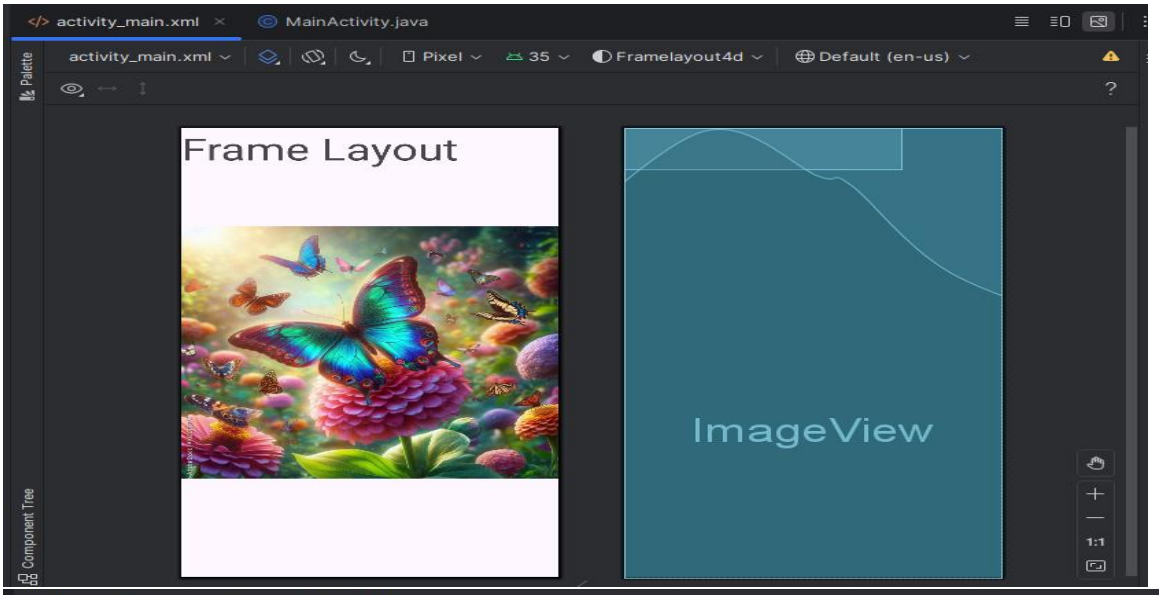
d. **Frame**

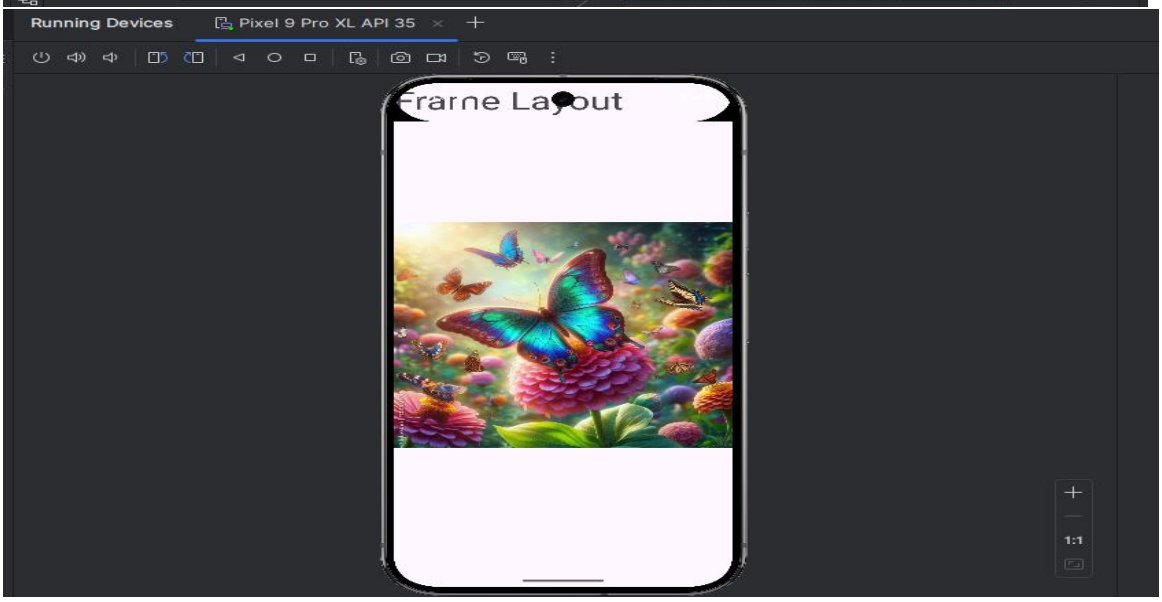
(MainActivity.java)

```
package com.example.framelayout4d;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
```

(activity_main.xml)

```xml
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" >
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="50sp"
        android:textAlignment="center"
        android:text="Frame Layout" />
    <ImageView
        android:layout_height="match_parent"
        android:layout_width="match_parent"
        android:src="@drawable/img" > </ImageView>
</FrameLayout>
```





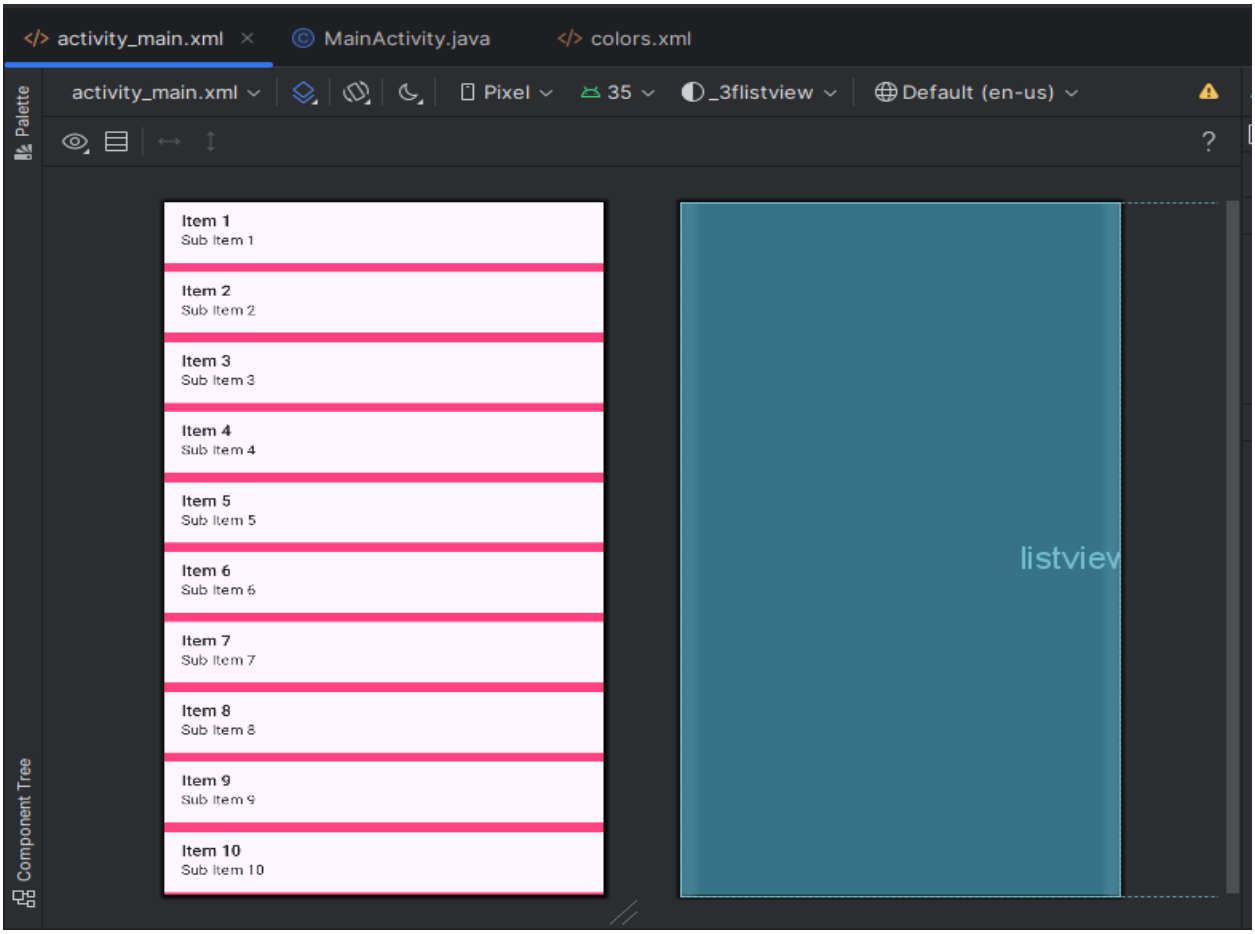
e. **List View**

(MainActivity.java)

```
package com.example.a3flistview;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;
import android.widget.ArrayAdapter;
import android.widget.ListView;
public class MainActivity extends AppCompatActivity {
    String[] mobileArray = {"Android","IPhone","WindowsMobile","Blackberry",
            "WebOS","Ubuntu","Windows7","Max OS X"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ArrayAdapter adapter = new ArrayAdapter<String>(this,
R.layout.activity_main, mobileArray);
        ListView listView = findViewById(R.id.listview1);
        listView.setAdapter(adapter);
    }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".list_view">
    <ListView
        android:id="@+id/listview1"
        android:layout_width="758dp"
        android:layout_height="match_parent"
        android:divider="@color/colorAccent"
        android:dividerHeight="10dp"
        android:outlineAmbientShadowColor="@color/colorAccent"></ListView>
</LinearLayout>
```



f. **Grid View**

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:columnCount="3"
    android:rowCount="3"
    android:orientation="horizontal">
    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="1" />
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="2" />
    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="3" />
```

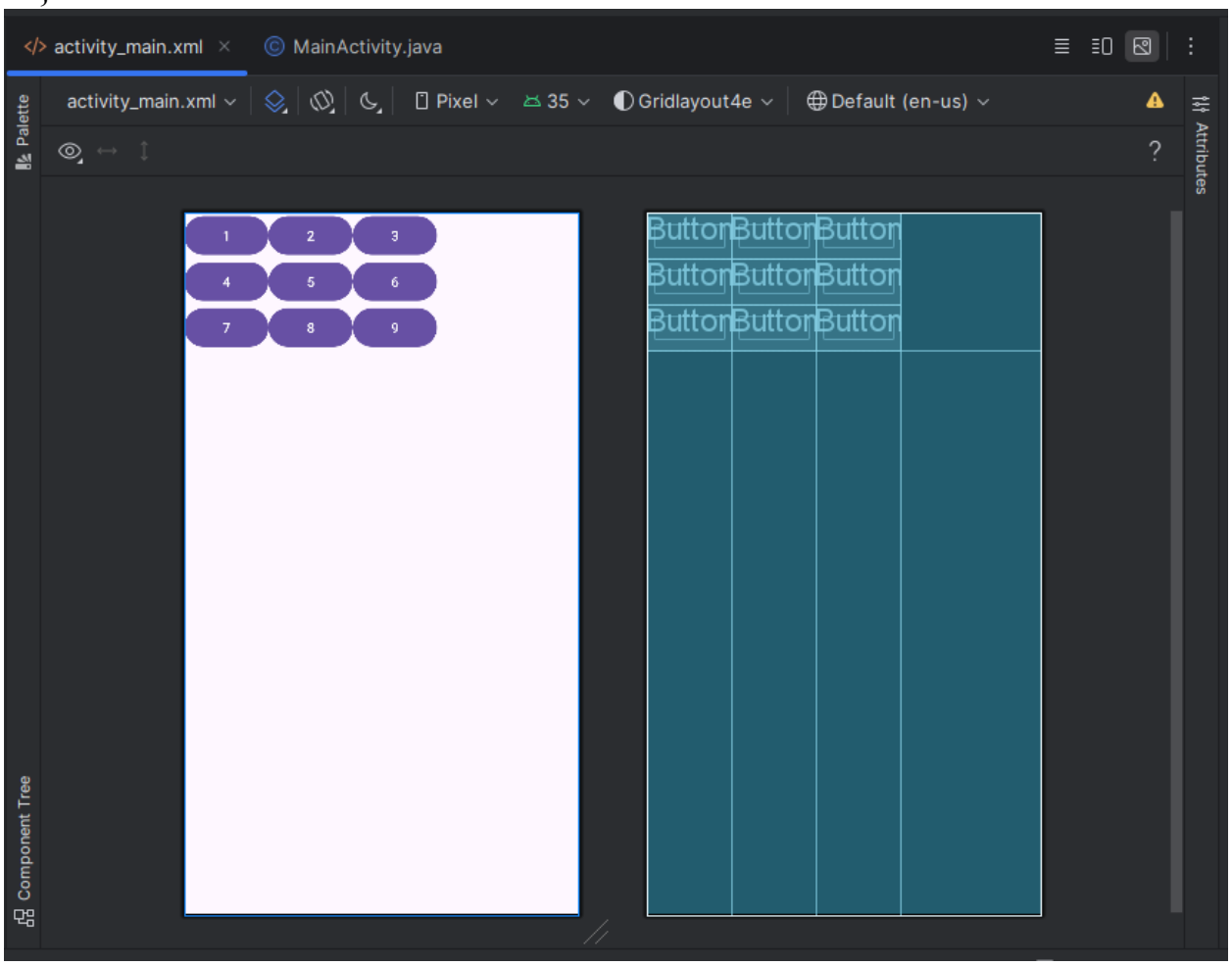
```
    <Button
        android:id="@+id/btn4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="4" />
    <Button
        android:id="@+id/btn5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="5" />
    <Button
        android:id="@+id/btn6"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="6" />
    <Button
        android:id="@+id/btn7"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="7" />
    <Button
        android:id="@+id/btn8"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="8" />
    <Button
        android:id="@+id/btn9"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="9" />
</GridLayout>
```

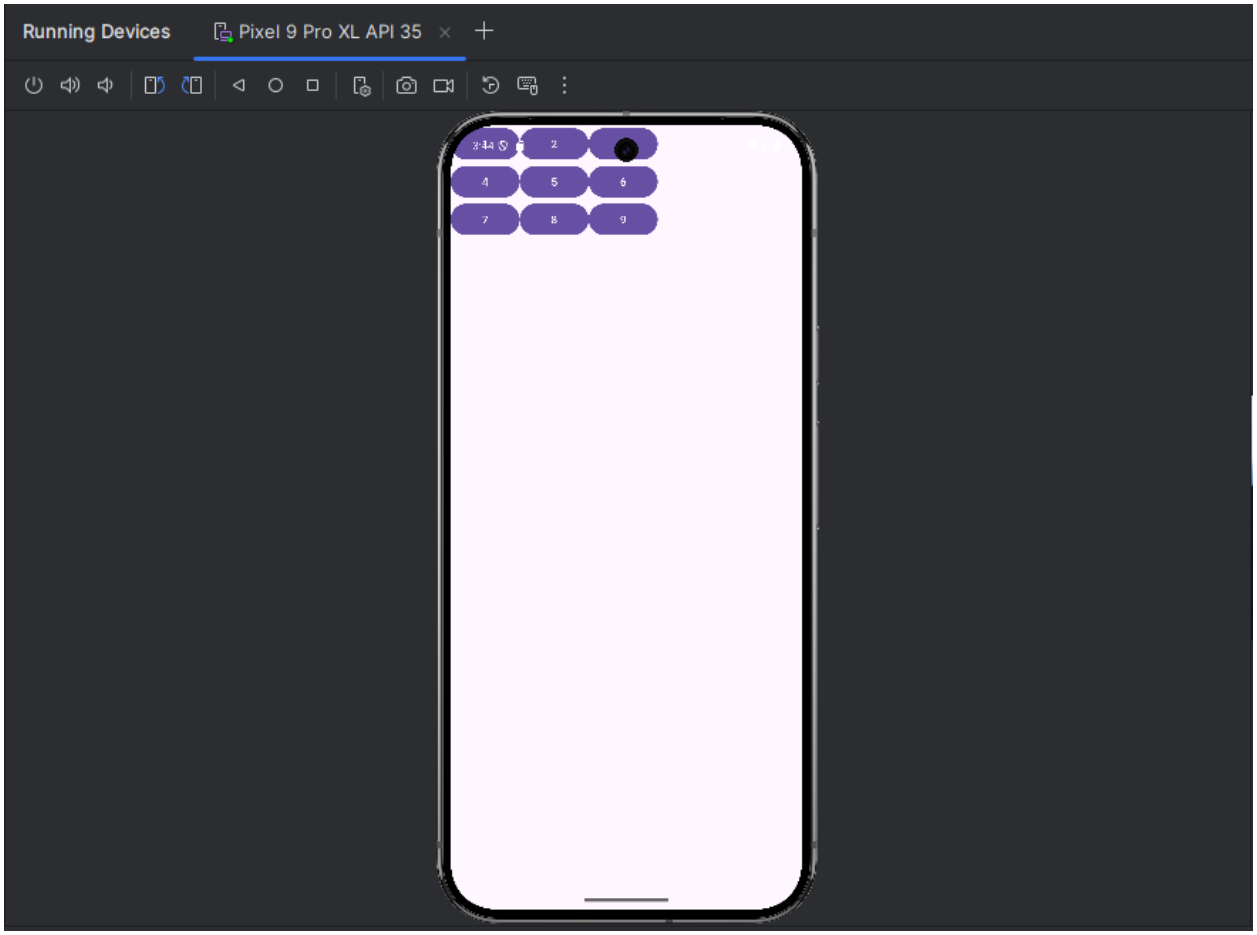
(MainActivity.java)

```
package com.example.gridlayout4e;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    private Button btnClick;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    btnClick = findViewById(R.id.btn1);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 1 clicked");
        }
    });
    btnClick = findViewById(R.id.btn2);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 2 clicked");
        }
    });
    btnClick = findViewById(R.id.btn3);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 3 clicked");
        }
    });
    btnClick = findViewById(R.id.btn4);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 4 clicked");
        }
    });
    btnClick = findViewById(R.id.btn5);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 5 clicked");
        }
    });
    btnClick = findViewById(R.id.btn6);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 6 clicked");
        }
    });
    btnClick = findViewById(R.id.btn7);
```

```
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 7 clicked");
        }
    });
    btnClick = findViewById(R.id.btn8);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 8 clicked");
        }
    });
    btnClick = findViewById(R.id.btn9);
    btnClick.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            showToast("Button 9 clicked");
        }
    });
}
private void showToast(String message) {
    Toast.makeText(getApplicationContext(), message, Toast.LENGTH_SHORT).show();
}
```




Running Devices
Pixel 9 Pro XL API 35
3:44
2
4
5
6
7
8
9

Practical:-4

Q] Programming UI element

AppBar, Fragments, UI Components

a. **AppBar**

- Create a new project, and go to:

ProjectName > App > Src > Main > Res > values > styles.xml Change the default line to this:

(Styles.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
   <style name="AppTheme" parent="Theme.AppCompat.Light.NoActionBar"></style>
</resources>
```

- Go to, ProjectName > App > Src > Main > Res > Layout
- Right Click on layout and add a new file “toolbar.xml”
- Go to toolbar.xml and change the default layout with this:

(toolbar.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.appcompat.widget.Toolbar
xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:background="@color/design_default_color_secondary"
   android:elevation="4dp">
</androidx.appcompat.widget.Toolbar>
```

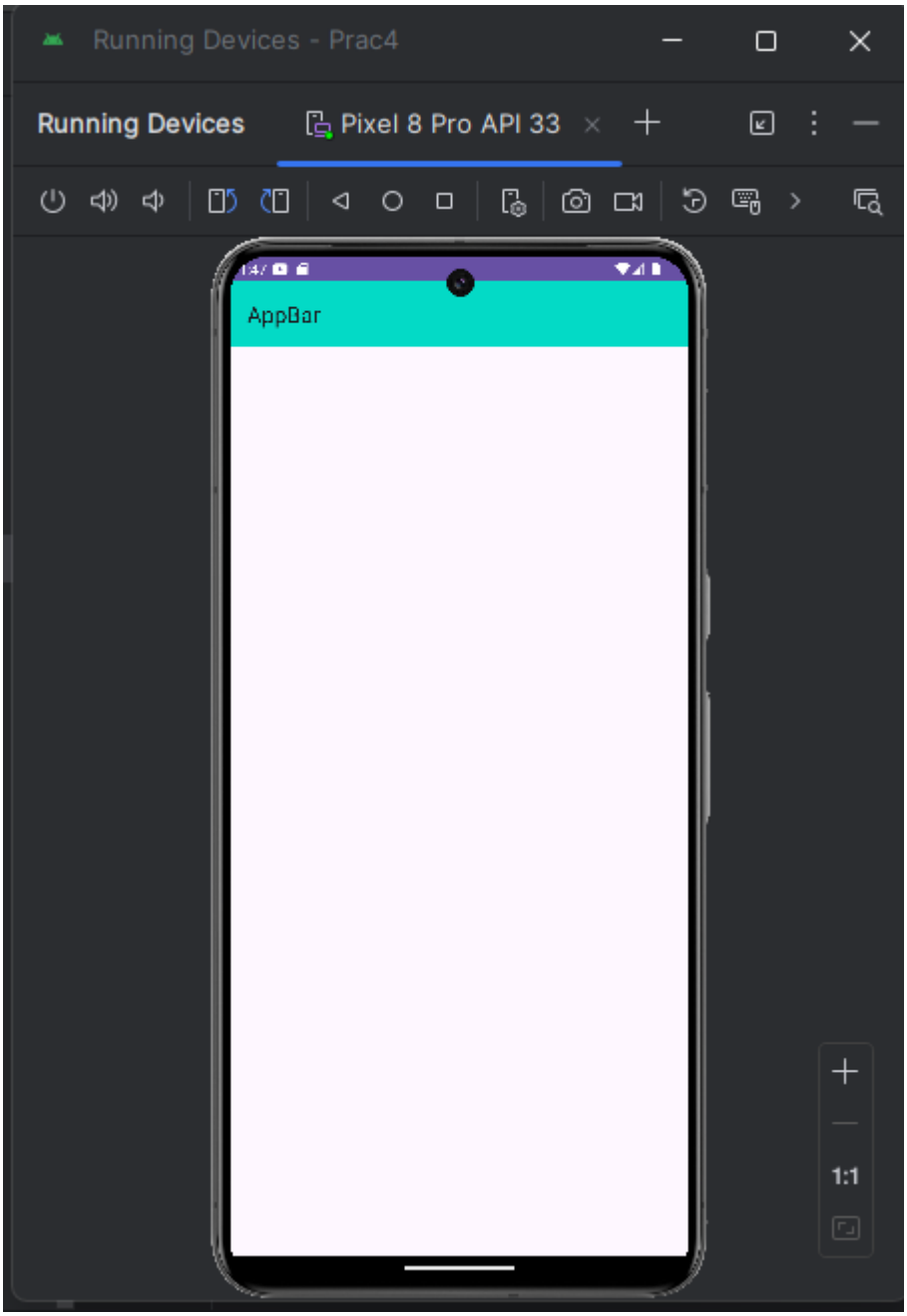
(MainActivity.java)

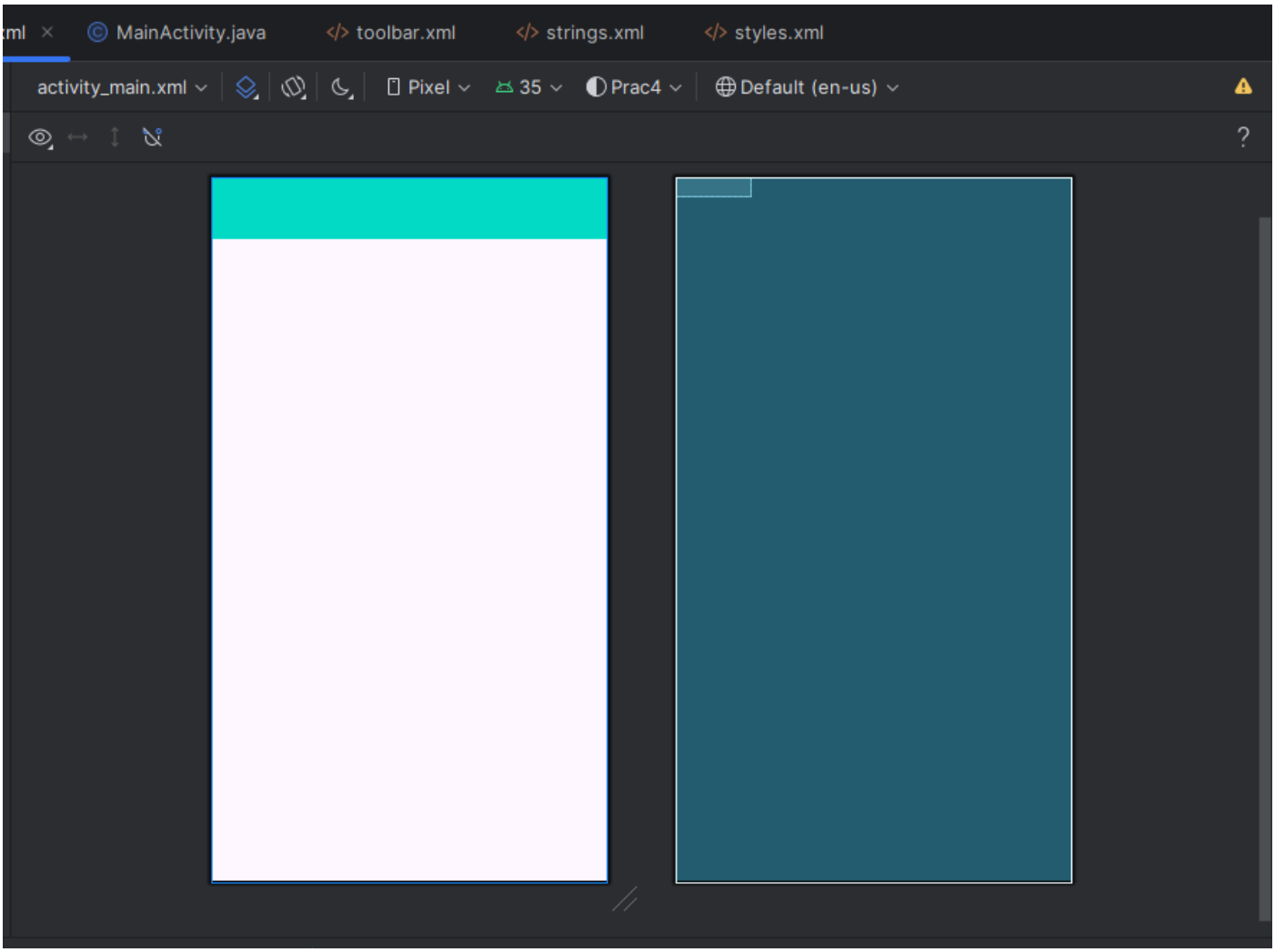
```
package com.example.prac4;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import androidx.appcompat.widget.Toolbar;
public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      Toolbar tbar =findViewById(R.id.toolbar);
      setSupportActionBar(tbar);
   }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
```

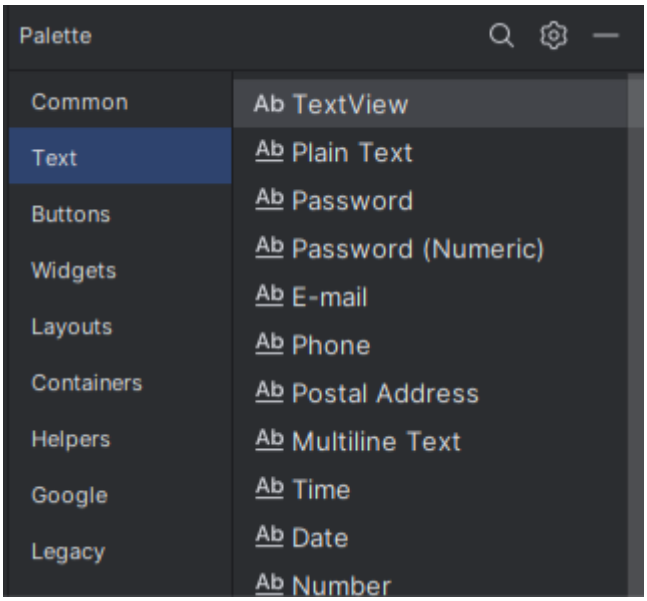
```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"/>
        <include
            android:id="@+id/toolbar"
            layout="@layout/toolbar"/>
</RelativeLayout>
```





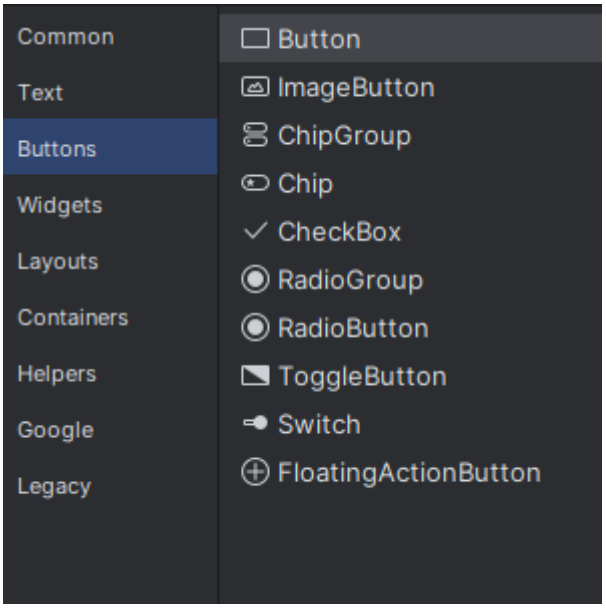
b. **Demonstration of UI Components(TextView, EditText, Button)**

TextView :Label Field In Android, TextView displays text to the user and optionally allows them to edit it programmatically. TextView is a complete text editor, however basic class is configured to not allow editing but we can edit it.



- EditText: Input Field In Android, EditText is a standard entry widget in android apps. It is an overlay over TextView that configures itself to be editable. EditText is a subclass of TextView with text editing operations. We often use EditText in our applications in order to provide an input or text field, especially in forms. The most simple example of EditText is Login or Sign-in form.

- Button: In Android, Button represents a push button. A Push buttons can be clicked, or pressed by the user to perform an action. There are different types of buttons used in android such as CompoundButton, ToggleButton, RadioButton.



Calculator Application
(MainActivity.java)

```
package com.example.a4aappbar;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.EditText;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    Button b1,b2,b3,b4;
    EditText t1,t2,t3;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

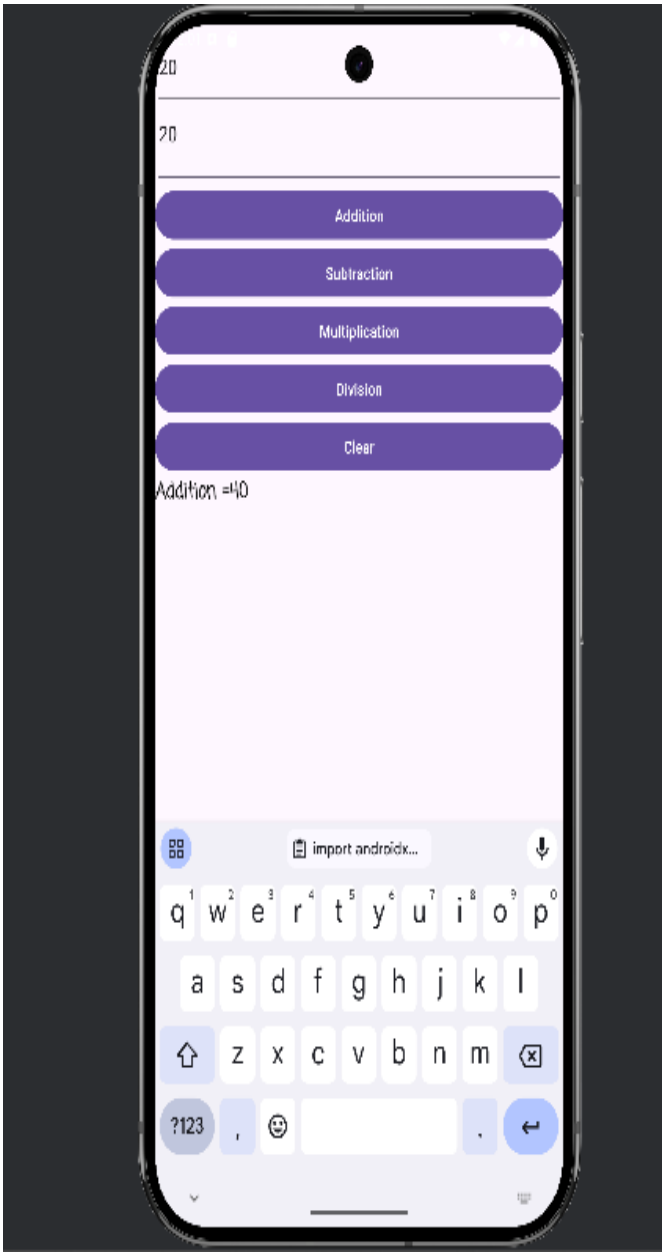
```
        t1=(EditText)findViewById(R.id.no1);
        t2=(EditText)findViewById(R.id.no2);
        t3=(EditText)findViewById(R.id.result);
        b1=(Button)findViewById(R.id.btnAdd);
        b2=(Button)findViewById(R.id.btnSub);
        b3=(Button)findViewById(R.id.btnDiv);
        b4=(Button)findViewById(R.id.btnMul);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Integer a=Integer.parseInt(t1.getText().toString());
                Integer b=Integer.parseInt(t2.getText().toString());
                Integer c=a+b;
                t3.setText(c.toString());
            }
        });
        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Integer a=Integer.parseInt(t1.getText().toString());
                Integer b=Integer.parseInt(t2.getText().toString());
                Integer c=a-b;
                t3.setText(c.toString());
            }
        });
        b3. setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Integer a=Integer.parseInt(t1.getText().toString());
                Integer b=Integer.parseInt(t2.getText().toString());
                Integer c=a/b;
                t3.setText(c.toString());
            }
        });
        b4.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Integer a=Integer.parseInt(t1.getText().toString());
                Integer b=Integer.parseInt(t2.getText().toString());
                Integer c=a*b;
                t3.setText(c.toString());
            }
        });
    }
}
```

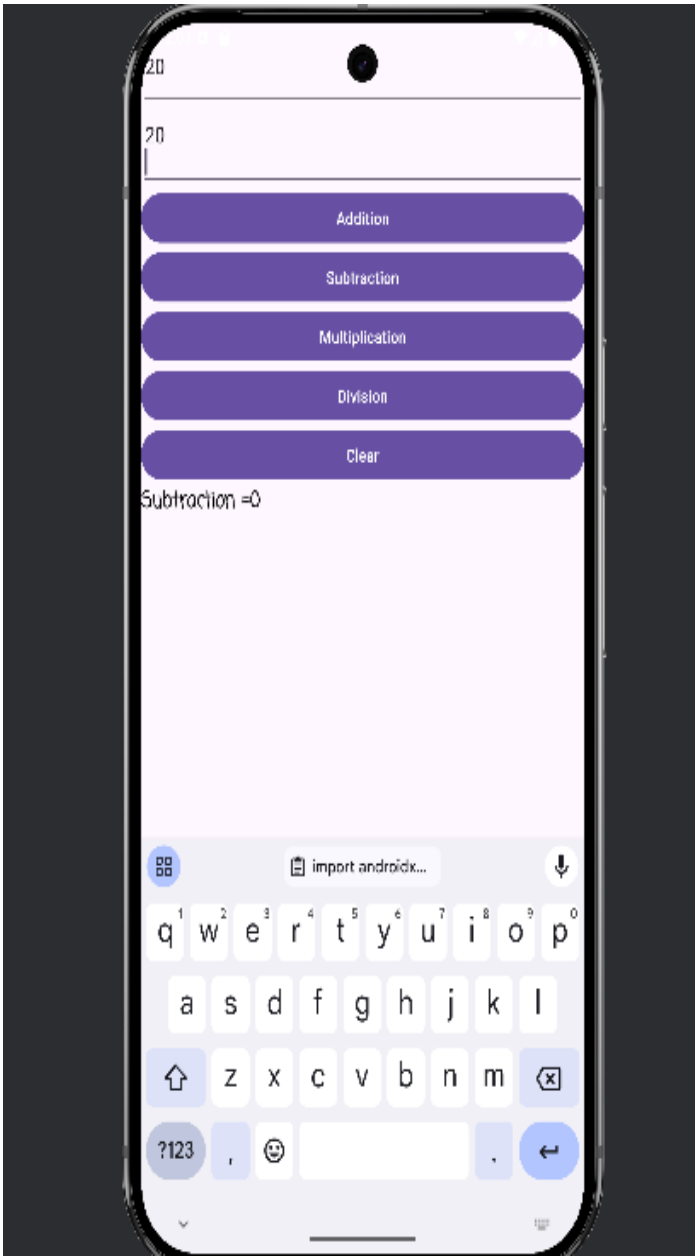
(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="@drawable/img"
tools:context=".CalculatorActivity">
   <EditText
      android:id="@+id/no1"
      android:layout_width="368dp"
      android:layout_height="68dp"
      android:layout_alignParentStart="true"
      android:layout_alignParentLeft="true"
      android:layout_alignParentTop="true"
      android:layout_centerHorizontal="true"
      android:layout_marginStart="64dp"
      android:layout_marginLeft="64dp"
      android:layout_marginTop="154dp"
      android:ems="10"
      android:hint="Enter no. 1"
      android:inputType="textPersonName"
      android:textColorHint="@color/colorPrimaryDark" />
   <EditText
   android:id="@+id/no2"
   android:layout_width="368dp"
   android:layout_height="65dp"
   android:layout_alignParentTop="true"
   android:layout_centerHorizontal="true"
   android:layout_marginTop="224dp"
   android:ems="10"
   android:hint="Enter no. 2"
   android:textColorHint="@color/colorPrimaryDark"
   android:inputType="textPersonName" />
<EditText
   android:id="@+id/result"
   android:layout_width="366dp"
   android:layout_height="68dp"
   android:layout_alignParentTop="true"
   android:layout_centerHorizontal="true"
   android:layout_marginTop="300dp"
   android:textColorHint="@color/colorPrimaryDark"
   android:ems="10"
   android:hint="Store Result"
   android:inputType="textPersonName" />
   <Button
```

```
    android:id="@+id/btnAdd"
    android:layout_width="141dp"
    android:layout_height="66dp"
    android:layout_alignParentStart="true"
    android:layout_alignParentLeft="true"
    android:layout_alignParentBottom="true"
    android:layout_marginStart="56dp"
    android:layout_marginLeft="56dp"
    android:layout_marginBottom="192dp"
    android:background="#6666"
    android:text="ADD(+)" />
<Button
    android:id="@+id/btnSub"
    android:layout_width="146dp"
    android:layout_height="70dp"
    android:layout_alignParentStart="true"
    android:layout_alignParentLeft="true"
    android:layout_alignParentBottom="true"
    android:layout_centerHorizontal="true"
    android:layout_marginStart="271dp"
    android:layout_marginLeft="271dp"
    android:layout_marginBottom="190dp"
    android:background="#6666"
    android:text="SUB(-)" />
<Button
    android:id="@+id/btnMul"
    android:layout_width="148dp"
    android:layout_height="69dp"
    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="70dp"
    android:layout_marginRight="70dp"
    android:layout_marginBottom="84dp"
    android:background="#6666"
    android:text="MUL(*)" />
<Button
    android:id="@+id/btnDiv"
    android:layout_width="160dp"
    android:layout_height="71dp"
    android:layout_alignParentStart="true"
    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    android:layout_alignParentBottom="true"
    android:layout_centerHorizontal="true"
    android:layout_marginStart="42dp"
    android:layout_marginEnd="290dp"
```

```
        android:layout_marginRight="290dp"
        android:layout_marginBottom="89dp"
        android:background="#6666"
        android:text="DIV(/)" />
</RelativeLayout>
```





Practical:-5

Q] Programming menus, dialog, dialog fragments

a. Creation of Menu

- Create a new project and go to,
- Project > App -> src -> res(Right-click on res and select) -> New -> Android Resource File -> Filename(menus) and Resource Type Menu
- Now go to, menus.xml file and make the following changes to code file,

(Menu.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:android="http://schemas.android.com/apk/res/android">
     <item
        android:id="@+id/new_id"
        android:enabled="false"
        android:title="New"
        android:visible="false" />
     <item
        android:id="@+id/save_id"
        android:enabled="true"
        android:title="Save"
        android:visible="true" />
     <item
        android:id="@+id/saveas_id"
        android:title="Save As"
        />
     <item
        android:id="@+id/about_id"
        android:title="About Us"
        />
     <item
        android:id="@+id/more_id"
        android:title="More Option"
        />
</menu>
```

(MainActivity.java)

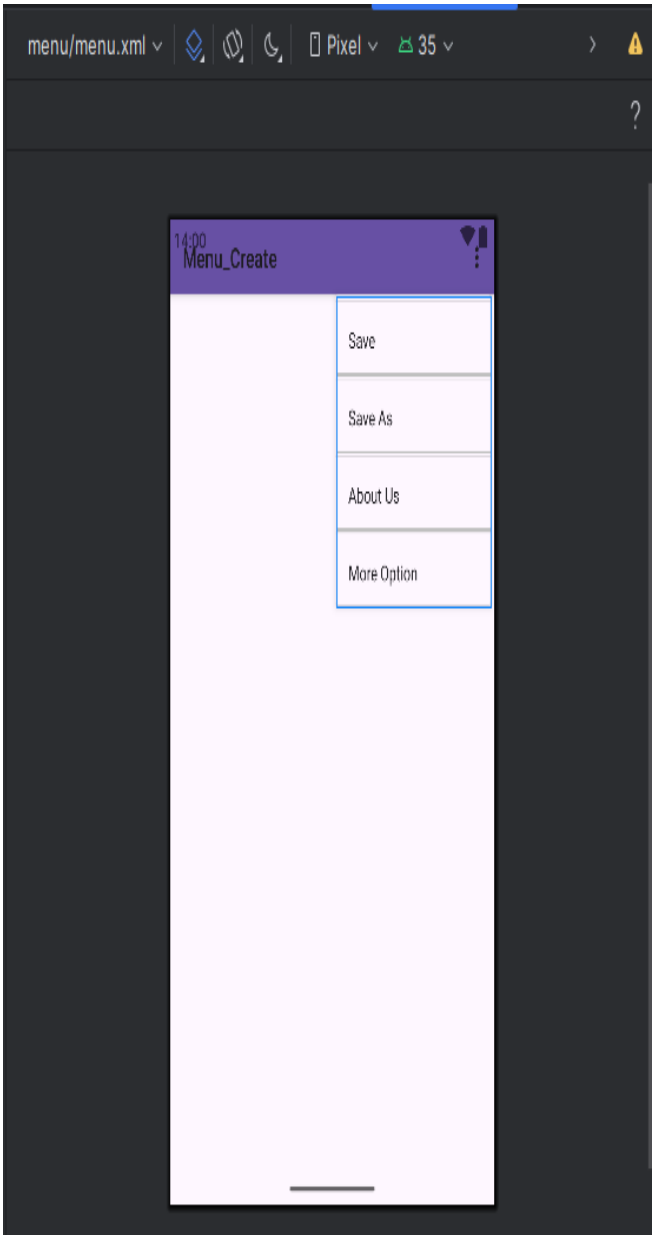
```
package com.example.menu_create;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.widget.Toolbar;
```

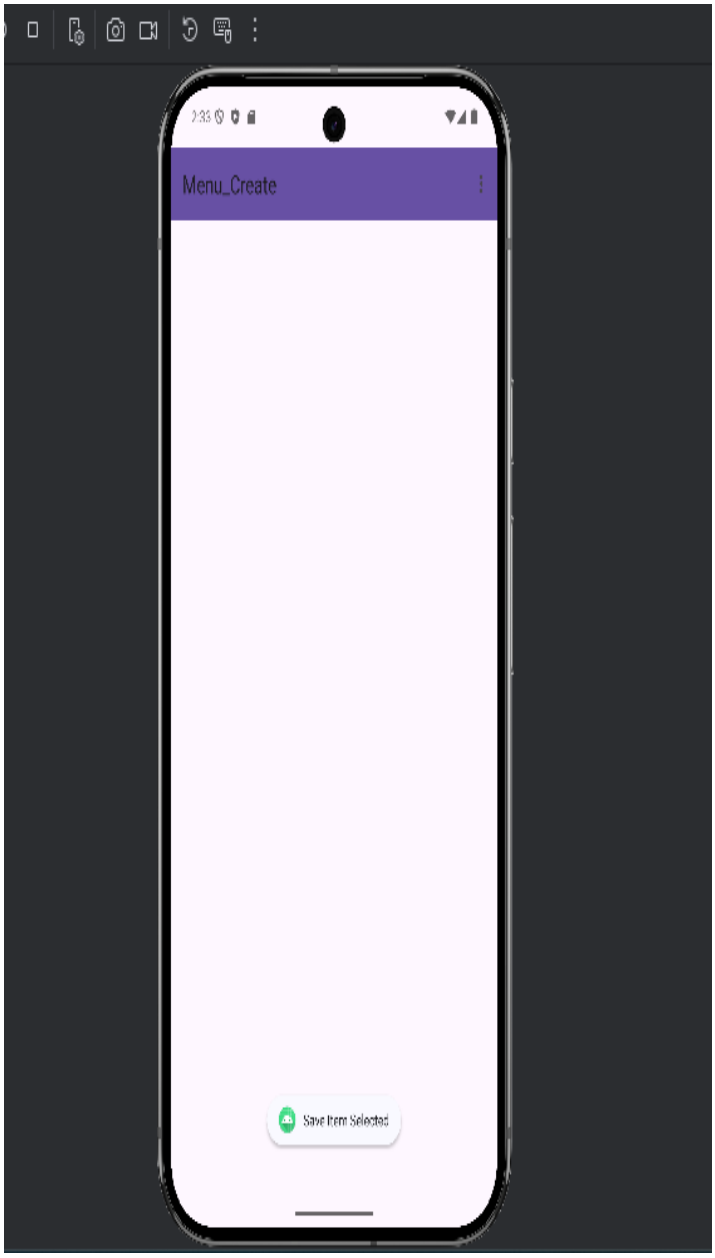
```
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
@Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        Toolbar tool = findViewById(R.id.toolbar);
        setSupportActionBar(tool);
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        MenuInflater m = getMenuInflater();
        m.inflate(R.menu.menu,menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int itemid = item.getItemId();
        if(itemid == R.id.new_id){
            Toast.makeText(this, "New Item Selected", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if(itemid == R.id.save_id){
            Toast.makeText(this, "Save Item Selected", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if(itemid == R.id.saveas_id){
            Toast.makeText(this, " Save As Item Selected", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if(itemid == R.id.more_id){
            Toast.makeText(this, "More option Item Selected", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if(itemid == R.id.about_id){
                Toast.makeText(this, "About Us Item Selected", Toast.LENGTH_SHORT).show();
```

```
            return true;
        }
        else{
            return super.onOptionsItemSelected(item);
        }
    }
}
```

(activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:background="?attr/colorPrimary"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```





b. Creation of Dialog

Dialogs in Android are pop-up windows that prompt the user to take action or provide information. They are used for interactions that require quick feedback or input from the user.There are different types of dialogs in Android, but they can generally be categorized into AlertDialog, ProgressDialog, DatePickerDialog, and TimePickerDialog. Among them, AlertDialog is the most used.

- Create a new project and
- Add the following code in

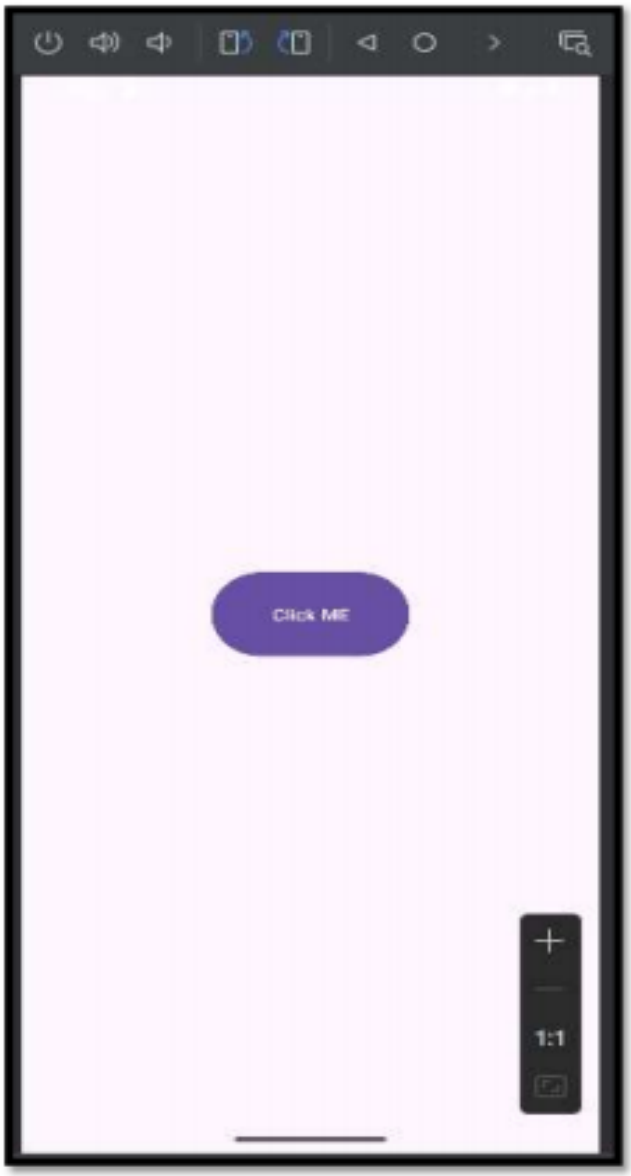
(MainActivity.java)

```
package com.example.alertdialogbox;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;
import android.content.DialogInterface;
import android.content.Intent;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Toast;
import android.os.Bundle;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
   Button btn;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      btn= findViewById(R.id.button);
      btn.setOnClickListener(new View.OnClickListener() {
                        @Override
                        public void onClick(View view) {
                           openAlertBox();
                        }
                     }
      );
   }
   private void openAlertBox() {
      AlertDialog.Builder builder = new
            AlertDialog.Builder(MainActivity.this);
      builder.setMessage("Do you want to exit ?");
      builder.setIcon(R.drawable.baseline_crisis_alert_24);
      builder.setTitle("Alert !");
      builder.setCancelable(false);
      builder.setPositiveButton("Yes", (DialogInterface.OnClickListener)
            (dialog, which) -> {
               // When the user click yes button then app will close
               finish();
builder.setNegativeButton("No", (DialogInterface.OnClickListener)
            (dialog, which) -> {
               // If user click no then dialog box is canceled.
               dialog.cancel();
            });
      // Create the Alert dialog
      AlertDialog alertDialog = builder.create();
      // Show the Alert Dialog box
      alertDialog.show();
   }
}
```

(activity_main.xml)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/button"
        android:layout_width="141dp"
        android:layout_height="79dp"
        android:text="EXIT" />
</LinearLayout>
```



Practical:-6

Q] Programs on Intents, Events, Listeners and Adapters

The Android Intent Class, Using Events and Event Listeners

In Android, Intents are used for communication between components (e.g., starting activities or services). Events trigger actions (like clicks or touch), while Event Listeners handle these events, executing code in response. Adapters bridge data sources and UI elements (e.g., ListView or RecyclerView), displaying data dynamically.

- Create a new project
- Now we need to create another file, newpage_layout. For this go to, ProjectName > App > src > main > res > layout > New > Layout Resource File > Name the file as newpage_layout > Click OK
- Now we have to create an Empty Activity. For creating an Empty Activity, go to, ProjectName > App > src > main > java > New > Activity > Empty Activity > Name the file as newpage > Finish

(MainActivity.java)

```
package com.example.intentpassingdataapp;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent; import android.graphics.Color; import android.os.Bundle;
import android.view.View; import android.widget.Button; import android.widget.EditText;
import android.widget.QuickContactBadge;
import android.widget.TextView;
import android.widget.Toast;
import java.util.Random;
public class MainActivity extends AppCompatActivity {
   EditText etFName; EditText etLName; Button btnSubmit;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
      etFName = (EditText) findViewById(R.id.etFName);
      etLName = (EditText) findViewById(R.id.etLName);
      btnSubmit = (Button) findViewById(R.id.btnSubmit);
      btnSubmit.setOnClickListener(new View.OnClickListener() { @Override
      public void onClick(View v) {
         Intent intent = new Intent(getApplicationContext(), ViewActivity.class);
intent.putExtra("fname", etFName.getText().toString()); intent.putExtra("lname",
etLName.getText().toString()); startActivity(intent);
      }
```

```
        });
    }}
```

(ViewActivity.java)

```
package com.example.intentpassingdataapp;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent; import android.os.Bundle; import android.widget.TextView;
public class ViewActivity extends AppCompatActivity { TextView tvView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState); setContentView(R.layout.activity_view);
        tvView = (TextView) findViewById(R.id.tvView); Intent intent = getIntent();
        String fName = intent.getStringExtra("fname"); String lName =
intent.getStringExtra("lname");
        tvView.setText("Your name is: " + fName + " " + lName);
    }
}
```

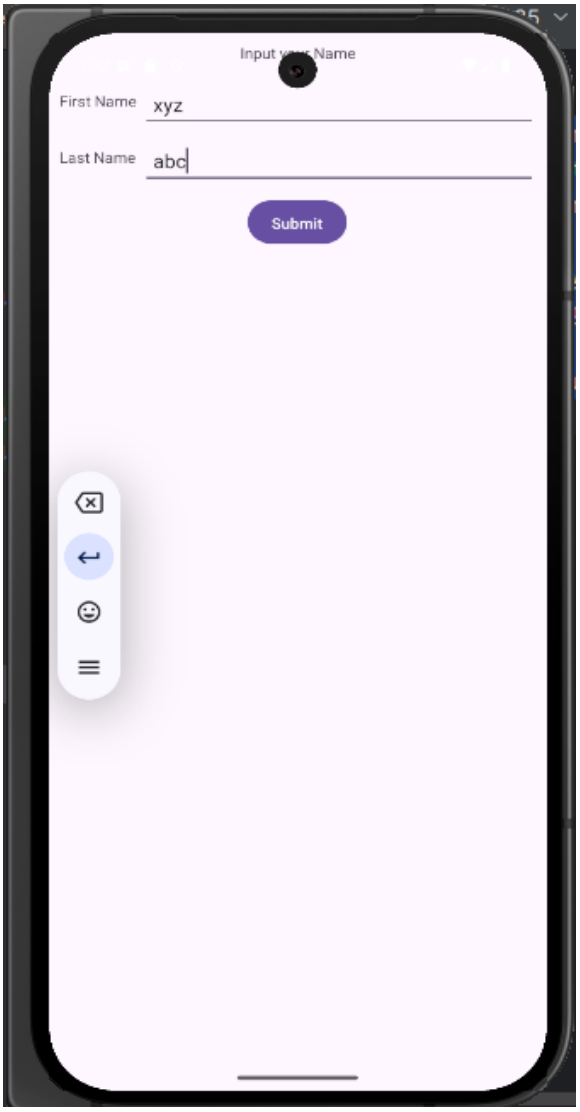
(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
android:layout_height="match_parent"
    tools:context=".MainActivity">
    <LinearLayout android:layout_width="match_parent"
android:layout_height="match_parent" android:orientation="vertical">
        <TextView android:layout_width="fill_parent" android:layout_height="wrap_content"
android:layout_marginTop="10dp" android:gravity="center_horizontal"
            android:text="Input your Name"></TextView>
        <TableLayout android:id="@+id/tableLayout1"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:layout_margin="10dp" android:stretchColumns="1">
            <TableRow android:id="@+id/tableRow1" android:layout_width="match_parent"
                android:layout_height="wrap_content">
                <TextView android:id="@+id/textView1"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content" android:text="First
Name"></TextView>
                <EditText android:id="@+id/etFName"
                    android:layout_width="match_parent" android:layout_height="48dp"
```

```
android:layout_marginLeft="5dp" android:layout_marginTop="5dp"
                android:hint="Enter your first name"
                android:labelFor="@id/textView1"
                android:padding="10dp"
                android:importantForAccessibility="yes">
                <requestFocus></requestFocus>
            </EditText>
        </TableRow>
        <TableRow android:id="@+id/tableRow2" android:layout_width="match_parent"
            android:layout_height="wrap_content">
            <TextView android:id="@+id/textView2"
                android:layout_width="match_parent" android:layout_height="wrap_content"
android:text="Last Name"></TextView>
            <EditText android:id="@+id/etLName"
                android:layout_width="match_parent" android:layout_height="48dp"
android:layout_marginLeft="5dp" android:layout_marginTop="5dp"
                android:hint="Enter your last name"
                android:labelFor="@id/textView2"
                android:padding="10dp"
                android:importantForAccessibility="yes"></EditText>
        </TableRow>
    </TableLayout>
    <Button android:id="@+id/btnSubmit"
        android:layout_width="wrap_content" android:layout_height="48dp"
android:layout_gravity="center_horizontal" android:text="Submit"
        android:padding="12dp"></Button>
  </LinearLayout>
</RelativeLayout>
```

(Activity_view.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
android:layout_height="match_parent"
  tools:context=".ViewActivity">
  <TextView android:id="@+id/tvView" android:layout_width="400dp"
android:layout_height="85dp"
      android:layout_gravity="center_horizontal" android:layout_marginTop="20dp"
android:text="Output" android:textColor="@android:color/holo_blue_dark"
android:textSize="24sp" android:textStyle="bold"></TextView>
</LinearLayout>
```


Input your Name
First Name
xyz
Last Name
abc
Submit

our name is: xyz abc

Practical:-7

Q] Programs on Services, notification and broadcast receivers

In Android, Services run in the background to perform long-running tasks, such as downloading files or playing music, without a UI. Notifications are used to inform users about important events, even when the app is not in the foreground. Broadcast Receivers listen for system-wide or app-specific events, like battery status changes or incoming messages, and respond by triggering specific actions. Together, these components enable apps to function efficiently, interact with users, and handle background processes effectively.

a. Services

- Create a new project
- Right-click on the java package > New > Java Class, name it MyService.
- Declare the Service in AndroidManifest.xml:
- Start the Service

(MainActivity.java)

```
package com.example.ringtone;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity { MediaPlayer mediaPlayer;
   ImageButton ibtn; @Override
   protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
      mediaPlayer = MediaPlayer.create(this,R.raw.myaudio);
   }
   public void playAudio(View v)
   {
      ibtn = findViewById(R.id.imageButton5);
      ibtn.setVisibility(View.VISIBLE);
      mediaPlayer.start();
      Toast.makeText(this, "Music is playing, Connect Ear Phone to Listen the Ringtone",
Toast.LENGTH_SHORT).show();
   }
   public void onPause(View v)
   {
```

```
        super.onPause(); mediaPlayer.release(); ibtn.setVisibility(View.INVISIBLE);
    }
}
```

(Activity_main.xml)

```
<Button android:id="@+id/btnPlay" android:layout_width="197dp" android:layout_height="75dp"
    android:layout_alignParentEnd="true" android:layout_alignParentRight="true"
android:layout_alignParentBottom="true" android:layout_marginEnd="87dp"
android:layout_marginRight="87dp" android:layout_marginBottom="468dp" android:text="Play
Audio"
    android:onClick="playAudio"
    />
<Button android:id="@+id/btnStop" android:layout_width="208dp" android:layout_height="70dp"
    android:layout_alignParentEnd="true" android:layout_alignParentRight="true"
android:layout_alignParentBottom="true" android:layout_marginEnd="89dp"
android:layout_marginRight="89dp" android:layout_marginBottom="356dp" android:text="Stop
Audio"
    android:onClick="onPause"/>
<ImageView
    android:id="@+id/imageView" android:layout_width="wrap_content"
android:layout_height="wrap_content" app:srcCompat="@color/red" />
<ImageButton android:id="@+id/imageButton5" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_alignParentEnd="true"
android:layout_alignParentRight="true" android:layout_alignParentBottom="true"
android:layout_marginEnd="66dp" android:layout_marginRight="66dp"
android:layout_marginBottom="182dp"
    android:visibility="invisible" app:srcCompat="@drawable/img" tools:visibility="invisible" />
```

b.Notification
(MainActivity.java)

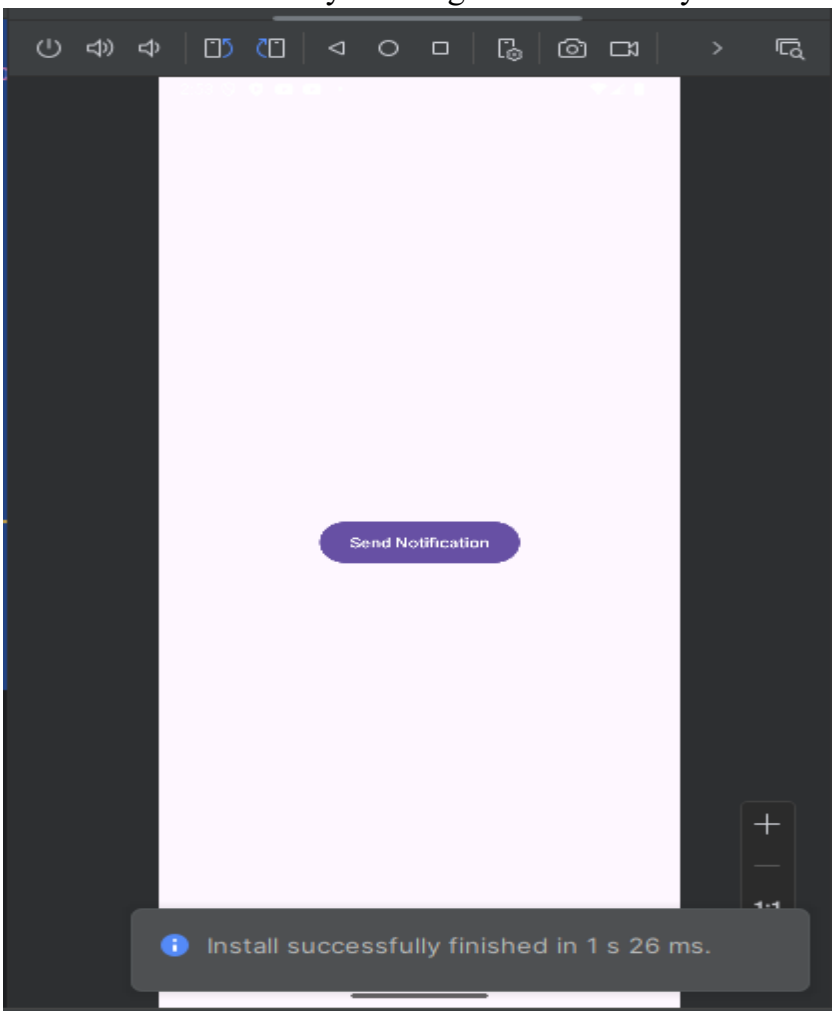
```
package com.example.notificationapp;
import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.util.Log;
import android.widget.Button;
import android.os.Bundle;
import android.content.pm.PackageManager;
import android.content.Context;
import android.Manifest;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
public class MainActivity extends AppCompatActivity {
    private static final String CHANNEL_ID = "default_channel";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Check for notification permission on Android 13+
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (ContextCompat.checkSelfPermission(this,
Manifest.permission.POST_NOTIFICATIONS)
                    != PackageManager.PERMISSION_GRANTED) {
                ActivityCompat.requestPermissions(this,
                        new String[]{Manifest.permission.POST_NOTIFICATIONS}, 1);
            }
        }
        // Create Notification Channel for Android 8.0 (API level 26) and above
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "Default Channel";
            String description = "This is the default notification channel";
            int importance = NotificationManager.IMPORTANCE_DEFAULT;
            NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name,
importance);
            channel.setDescription(description);
            NotificationManager notificationManager = getSystemService(NotificationManager.class);
            notificationManager.createNotificationChannel(channel);
        }
        // Get reference to the button and set an OnClickListener
        Button notifyButton = findViewById(R.id.notifyButton);
        notifyButton.setOnClickListener(v -> sendNotification());
    }
```

```
    @Override
    public void onRequestPermissionsResult(int requestCode, String[] permissions, int[]
grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == 1) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                // Permission granted, now trigger the notification
                sendNotification();
            } else {
                Toast.makeText(this, "Permission denied to show notifications",
Toast.LENGTH_SHORT).show();
            }
        }
    }
    private void sendNotification() {
        // Log to see if this method is being triggered
        Log.d("NotificationTest", "sendNotification called");

        // Create a simple notification
        Notification notification = new Notification.Builder(this, CHANNEL_ID)
                .setContentTitle("Test Notification")
                .setContentText("This is a simple notification!")
                .setSmallIcon(android.R.drawable.ic_dialog_info)  // Use a default icon
                .build();
        // Send the notification
        NotificationManager notificationManager = (NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
        notificationManager.notify(1, notification);  // 1 is the notification ID
    }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Button to trigger notification -->
    <Button
        android:id="@+id/notifyButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:text="Send Notification"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toBottomOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```



c. **Broadcast**

(MainActivity.java)

```
package com.example.broadcast;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Find the button by its ID
        Button btnOpenActivity = findViewById(R.id.btnOpenActivity);
```

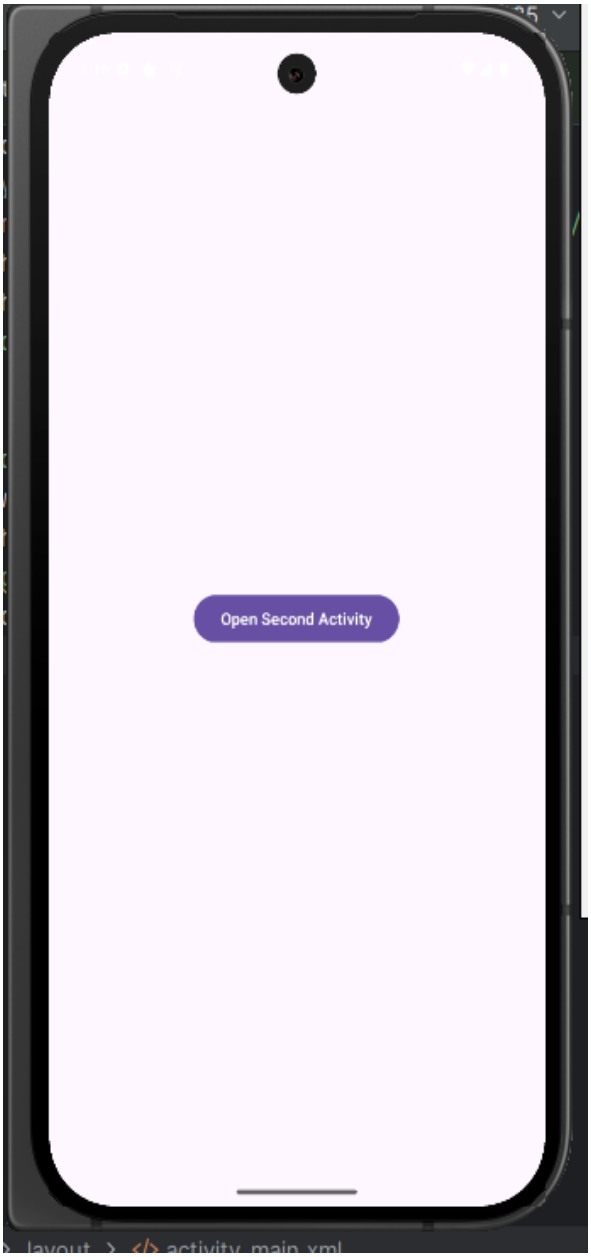
```
        // Set an OnClickListener (Event Listener) for the button
        btnOpenActivity.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Create an Intent to open SecondActivity
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                // Optionally, pass data to the second activity
                intent.putExtra("message", "Hello from MainActivity!");
                // Start the second activity
                startActivity(intent);
            }
        });
    }
}
```

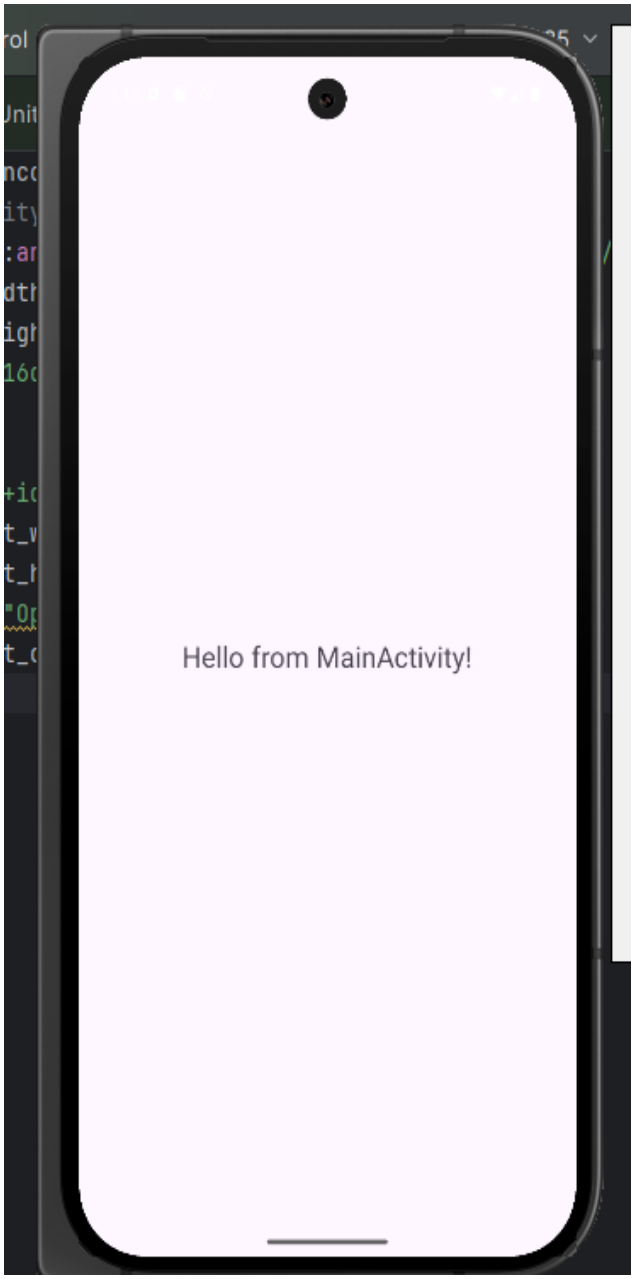
(SecondActivity.java)

```
package com.example.broadcast;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
        // Find the TextView by its ID
        TextView tvMessage = findViewById(R.id.tvMessage);
        // Get the intent that started this activity
        Intent intent = getIntent();
        // Check if the intent has extra data
        if (intent != null && intent.hasExtra("message")) {
            // Retrieve the message from the intent
            String message = intent.getStringExtra("message");
            // Display the message in the TextView
            tvMessage.setText(message);
        }
    }
}
```

(Activity_main.xml)

```xml
<?xml version="1.0" encoding="utf-8"?>
<!-- res/layout/activity_main.xml -->
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <Button
        android:id="@+id/btnOpenActivity"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Second Activity"
        android:layout_centerInParent="true"/>
</RelativeLayout>
```





Practical:-8

Q] Database Programming with SQLite

Database programming with SQLite in Android involves using SQLite to store and manage structured data locally within an app. It includes creating a database, defining tables, performing CRUD operations (Create, Read, Update, Delete), and using SQLiteOpenHelper to handle database creation and version management for efficient data storage and retrieval.

- Create a new project
- Right-click your java package > New > Java Class

(MainActivity.java)

```
package com.example.sqlite;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {
    mydb dbHelper; // Database helper
    EditText txt_name, txt_sal; // UI components
    Button btn_add, btn_del, btn_view, btn_update; // UI components
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        dbHelper = new mydb(this);
        txt_name = findViewById(R.id.txt_name);
        txt_sal = findViewById(R.id.txt_sal);
        btn_add = findViewById(R.id.btn_add);
        btn_del = findViewById(R.id.btn_del);
        btn_view = findViewById(R.id.btn_view);
        btn_update = findViewById(R.id.btn_update);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        btn_add.setOnClickListener(
                new View.OnClickListener() {
```

```
                @Override
                public void onClick(View v) {
                    String name = txt_name.getText().toString().trim();
                    String sala =txt_sal.getText().toString().trim();
                    int sal = Integer.parseInt(sala);
                    if (name.isEmpty() &&  sala.isEmpty()){
                        Toast.makeText(MainActivity.this, "Any field cannot be empty",
Toast.LENGTH_SHORT).show();
                    }
                    else{
                        dbHelper.insert_val(name,sal);
                        Toast.makeText(MainActivity.this, "Data inserted successfully",
Toast.LENGTH_SHORT).show();
                    }
                }
            }
        );
        btn_view.setOnClickListener(
            new View.OnClickListener() {

                @Override
                public void onClick(View v) {
                    String name = txt_name.getText().toString().trim();
                    String sala =txt_sal.getText().toString().trim();
                    int sal = Integer.parseInt(sala);
                    ArrayList<Colomn_Datatype> data = dbHelper.fetchdata();
                    if (data.isEmpty()) {
                        Toast.makeText(MainActivity.this, "No data found",
Toast.LENGTH_SHORT).show();
                    } else {
                        StringBuilder dataString = new StringBuilder();
                        for (Colomn_Datatype record : data) {
                            dataString.append("Name: ").append(record.Name).append(", Salary:
").append(record.salary).append("\n");
                        }
                        Toast.makeText(MainActivity.this, dataString.toString(),
Toast.LENGTH_LONG).show();
                    }
                }
            }
        );
        btn_update.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = txt_name.getText().toString().trim(); // Name of the record to update
                String sala = txt_sal.getText().toString().trim(); // New salary value
                if (name.isEmpty() || sala.isEmpty()) {
```

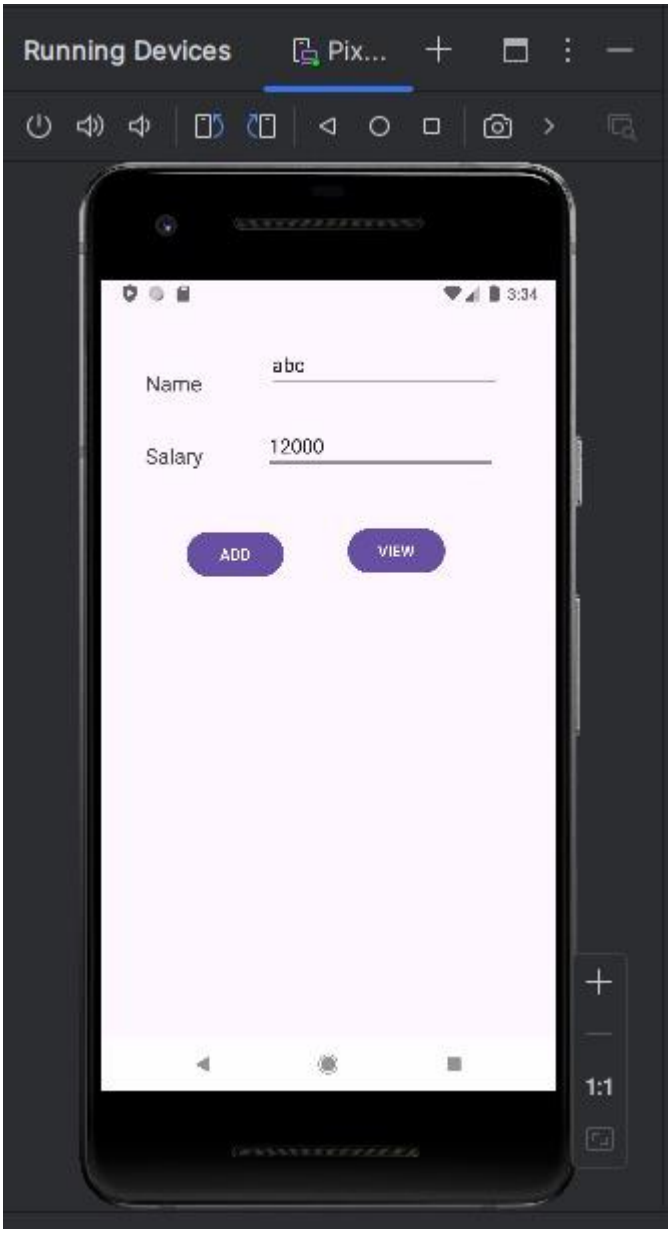
```
                Toast.makeText(MainActivity.this, "Name and Salary fields cannot be empty",
Toast.LENGTH_SHORT).show();
            } else {
                int sal = Integer.parseInt(sala); // Convert salary to integer
                boolean isUpdated = dbHelper.updateData(name, sal); // Call update method

                if (isUpdated) {
                    Toast.makeText(MainActivity.this, "Record updated successfully",
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Record not found or update failed",
Toast.LENGTH_SHORT).show();
                }
            }
        }
    });
    btn_del.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Get the name of the employee to delete
            String name = txt_name.getText().toString().trim();
            // Check if the name field is empty
            if (name.isEmpty()) {
                txt_name.setError("Please enter an employee name");
                return; // Stop further execution
            }
            // Call the delete function by name
            int rowsDeleted = dbHelper.deleteEmployeeByName(name);
            // Show a message based on the result
            if (rowsDeleted > 0) {
                Toast.makeText(MainActivity.this, "Employee deleted successfully",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this, "Employee not found or delete failed",
Toast.LENGTH_SHORT).show();
            }
        }
    });
  }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="40dp"
        android:layout_marginTop="56dp"
        android:text="Name"
        android:textSize="19sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <EditText
        android:id="@+id/txt_name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="60dp"
        android:layout_marginTop="30dp"
        android:ems="10"
        android:inputType="text"
        android:text="Name"
        android:hint=" Enter Name"
        app:layout_constraintStart_toEndOf="@+id/textView"
        app:layout_constraintTop_toTopOf="parent" />
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="40dp"
        android:layout_marginTop="40dp"
        android:text="Salary"
        android:textSize="19sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView" />
    <EditText
        android:id="@+id/txt_sal"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="56dp"
```

```
    android:layout_marginTop="28dp"
    android:ems="10"
    android:inputType="text"
    android:text="Name"
    android:hint=" Enter Salary"
    app:layout_constraintStart_toEndOf="@+id/textView2"
    app:layout_constraintTop_toBottomOf="@+id/txt_name" />
<Button
    android:id="@+id/btn_add"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="56dp"
    android:layout_marginTop="52dp"
    android:text="ADD"
    app:layout_constraintEnd_toStartOf="@+id/btn_view"
    app:layout_constraintHorizontal_bias="0.27"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2" />
<Button
    android:id="@+id/btn_view"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="48dp"
    android:layout_marginEnd="100dp"
    android:text="VIEW"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/txt_sal" />
<Button
    android:id="@+id/btn_del"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="76dp"
    android:layout_marginTop="24dp"
    android:layout_marginBottom="412dp"
    android:text="DELETE"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/btn_add" />
<Button
    android:id="@+id/btn_update"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="52dp"
    android:layout_marginTop="28dp"
    android:layout_marginBottom="411dp"
    android:text="UPDATE"
    app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintStart_toEndOf="@+id/btn_del"
        app:layout_constraintTop_toBottomOf="@+id/btn_view" />
</androidx.constraintlayout.widget.ConstraintLayout>
```



Practical:-9

Q] Programming threads, handles and asynchronized programs

In Android programming, threads run tasks in the background, preventing the UI from freezing. Handles are used to communicate between threads, like updating the UI from a background thread. Asynchronous programming, often using AsyncTask or LiveData, helps run tasks like network requests without blocking the main UI thread.

- Create a new project
- Add a TextView or Button to interact with or display results
- In MainActivity.java, create a new Thread to perform tasks in the background.
- Click Run in Android Studio to deploy the app to an emulator or physical devic

(MainActivity.java)

```
package com.example.prac9;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
 import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
public class MainActivity extends AppCompatActivity {
private TextView statusTextView;
private Button startButton;
private Handler handler = new Handler(Looper.getMainLooper()); // Handler for main thread private
ExecutorService executorService = Executors.newSingleThreadExecutor(); // Executor
for background tasks
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
statusTextView = findViewById(R.id.statusTextView);
 startButton = findViewById(R.id.startButton);
// Lambda expression for setting click listener
startButton.setOnClickListener(v -> {
// Submit a background task to the executor
executorService.submit(() -> {
// Simulating a long-running task
 for (int i = 1; i <= 5; i++) {
try {
Thread.sleep(1000); // Simulating work
} catch (InterruptedException e) {
e.printStackTrace();
}
// Store the value of i in a final variable to use inside the lambda
final int progress = i;
// Update UI on the main thread
runOnUiThread(() -> statusTextView.setText("Running task... " + progress));
```

```
}
// Final update after the task is complete
runOnUiThread(() -> statusTextView.setText("Task Completed!"));
});
});
}
@Override
protected void onDestroy() { super.onDestroy();
// Shutdown executor service when activity is destroyed executorService.shutdown();
}
}
```

(Activity_main.xml)

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".MainActivity">
<TextView android:id="@+id/statusTextView" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Status" android:textSize="18sp"
android:layout_marginTop="100dp"
app:layout_constraintTop_toTopOf="parent" app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"/>
<Button
android:id="@+id/startButton" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Start Task" android:layout_marginTop="50dp"
app:layout_constraintTop_toBottomOf="@id/statusTextView" app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Practical:-10

A] Programming Media API and Telephone API

The Media API in Android allows access to media features like audio, video, and image capture, enabling tasks such as recording, playback, and media file manipulation. The Telephone API provides functionality for managing phone calls, like initiating, receiving, or ending calls, and accessing call-related information.

- Create a new Android project in Android Studio.
- Add the necessary permissions in AndroidManifest.xml
- Use MediaPlayer for playback or MediaRecorder for recording.

a. Programming Media API

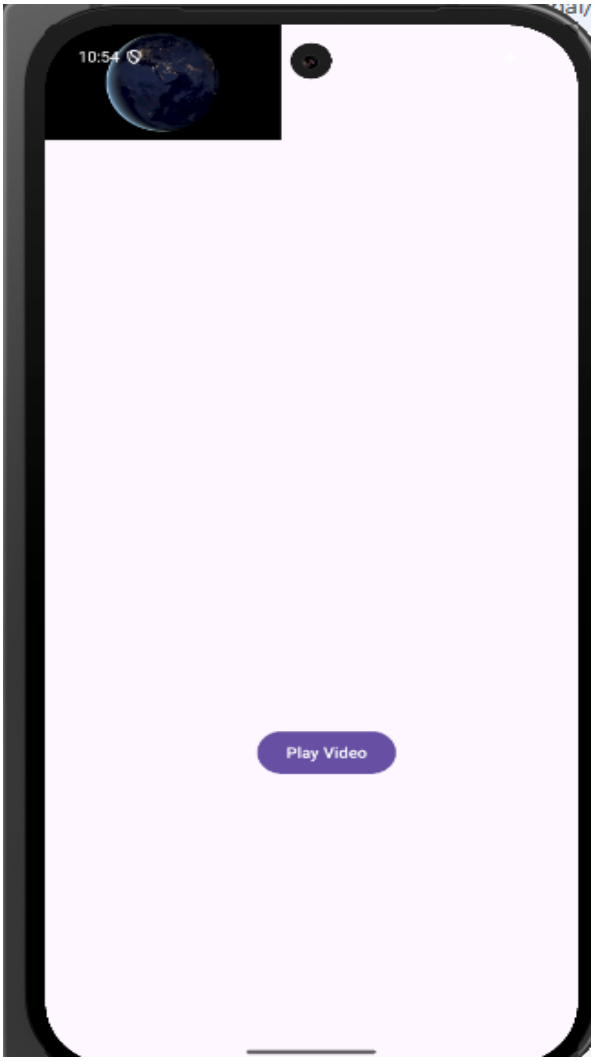
(MainActivity.java)

```
package com.example.playvedio;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.os.Environment;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.ImageButton; import android.widget.MediaController; import android.widget.Toast;
import android.widget.VideoView;
public class MainActivity extends AppCompatActivity { VideoView videoview;
   Button btn;
   @Override
   protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
   }
   public void playVideo(View view)
   {
      videoview = (VideoView) findViewById(R.id.videoView);
      try {
         MediaController mediacontroller = new MediaController( MainActivity.this);
         mediacontroller.setAnchorView(videoview);
         Uri uri = Uri.parse("android.resource://" + getPackageName() + "/" + R.raw.myvideo);
         videoview.setMediaController(mediacontroller); videoview.setVideoURI(uri);
      } catch (Exception e) { Log.e("Error", e.getMessage()); e.printStackTrace();
      }
      videoview.requestFocus();
      videoview.setOnPreparedListener(new MediaPlayer.OnPreparedListener() {
        public void onPrepared(MediaPlayer mp) {
            videoview.start();
         }
```

```
        });
    }}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
    tools:context=".MainActivity">
    <VideoView android:id="@+id/videoView" android:layout_width="wrap_content"
        android:layout_height="wrap_content" />
    <Button android:id="@+id/btnPlay"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_alignParentEnd="true" android:layout_alignParentRight="true"
android:layout_alignParentBottom="true" android:layout_marginEnd="151dp"
android:layout_marginRight="151dp" android:layout_marginBottom="274dp"
android:onClick="playVideo" android:text="Play Video" />
</RelativeLayout>
```



b. Telephone API

- Add the required permissions for phone functionality in AndroidManifest.xml
- Use the TelephonyManager to get phone state and manage calls
- To make a phone call programmatically

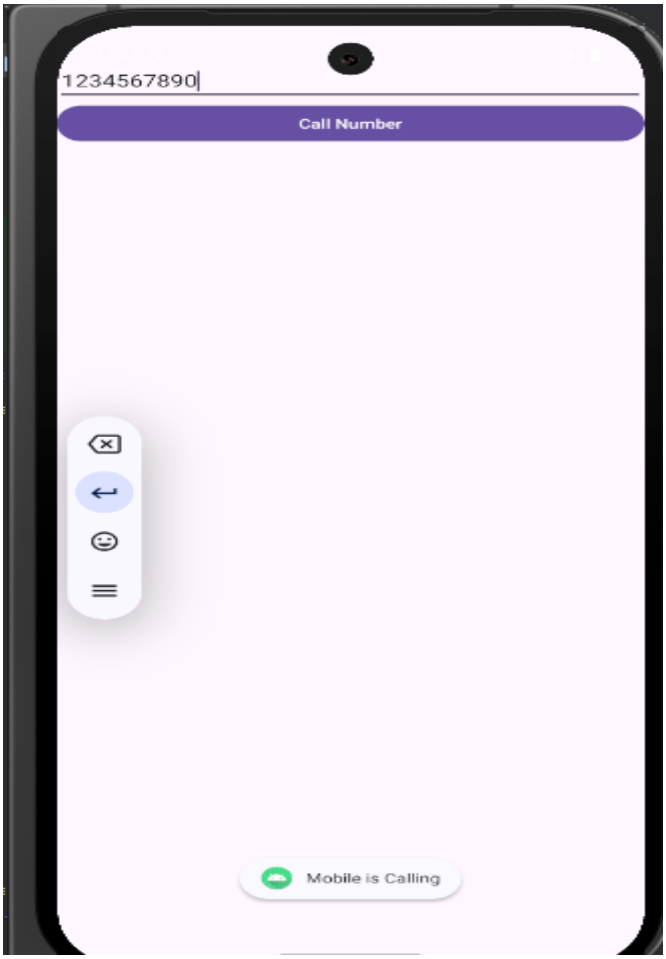
(MainActivity.java)

```
package com.example.telephoneapi;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.graphics.Color;
import android.os.Bundle;
import android.app.Activity; import android.content.Intent; import android.net.Uri; import
android.os.Bundle; import android.view.Menu; import android.view.View; import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity { @Override
protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
   Button dialPhNumber = (Button) findViewById(R.id.dialPhNumber);
dialPhNumber.setOnClickListener(new View.OnClickListener() {
      public void onClick(View view) {
         Toast.makeText(getApplicationContext(),"Mobile is Calling",Toast.LENGTH_SHORT).show();
         try {
            EditText phNumber = (EditText) findViewById(R.id.phNumber);
            Intent implicit1 = new Intent(Intent.ACTION_CALL, Uri.parse("tel:" +
phNumber.getText().toString()));
            startActivity(implicit1); Toast.makeText(getApplicationContext(),"Mobile is
Calling",Toast.LENGTH_SHORT).show();
         }
         catch (SecurityException e)
         {}
      }
   });
}
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
   tools:context=".MainActivity">
   <LinearLayout android:layout_width="match_parent" android:layout_height="match_parent"
android:orientation="vertical">
      <EditText
         android:hint="Enter Phone Number" android:id="@+id/phNumber"
```

```
android:layout_width="fill_parent" android:layout_height="81dp" android:ems="5" />
        <Button android:id="@+id/dialPhNumber" android:layout_width="fill_parent"
android:layout_height="45dp" android:text="Call Number" />
    </LinearLayout>
</RelativeLayout>
```



B] Programming Security and permissions

Programming security and permissions involve controlling access to software resources and data based on user roles and privileges. It ensures that only authorized users can access or modify certain parts of a system. Proper implementation prevents unauthorized actions, safeguarding the integrity and confidentiality of the system.

- Create a New Android Project
- Use Keystore for Secure Data Storage
- Use HTTPS for Network Communication
- Implement Secure Authentication

(MainActivity.java)

```
package com.example.login_page;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Bundle;
import android.view.View; import android.widget.Button; import android.widget.EditText; import
android.widget.TextView; import android.widget.Toast;
public class MainActivity extends AppCompatActivity { Button b1,b2;
   EditText ed1,ed2;
   TextView tx1; int counter = 3; @Override
   protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
      b1 = (Button)findViewById(R.id.button);
      ed1 = (EditText)findViewById(R.id.editText);
      ed2 = (EditText)findViewById(R.id.editText2);
      b2 = (Button)findViewById(R.id.button2);
      tx1 = (TextView)findViewById(R.id.textView3);
      b1.setOnClickListener(new View.OnClickListener() { @Override
      public void onClick(View v) {
         if(ed1.getText().toString().equals("admin") &&
               ed2.getText().toString().equals("admin")) { Toast.makeText(getApplicationContext(),
               "Redirecting...",Toast.LENGTH_SHORT).show();
         }else{
            Toast.makeText(getApplicationContext(), "Wrong
Credentials",Toast.LENGTH_SHORT).show();
            tx1.setVisibility(View.VISIBLE); tx1.setBackgroundColor(Color.RED); counter--;
tx1.setText(Integer.toString(counter));
            if (counter == 0) {
               b1.setEnabled(false);
            }
         }
      }
      });
      b2.setOnClickListener(new View.OnClickListener() { @Override
      public void onClick(View v) { finish();
      }
      });
   }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"android:layout_width="match_parent"
android:layout_height="match_parent"
   tools:context=".MainActivity">
    <TextView android:id="@+id/textview"
      android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_alignParentTop="true" android:layout_centerHorizontal="true" android:text="Login
Example" android:textSize="35dp" />
```

```
<!-- EditText for entering name -->
<EditText android:id="@+id/editText"
    android:layout_width="match_parent"
    android:layout_height="48dp"
    android:hint="Enter Name"
    android:focusable="true"
    android:textColorHighlight="#ff7eff15"
    android:layout_marginTop="46dp"
    android:layout_alignParentLeft="true"
    android:layout_alignParentStart="true"
    android:layout_alignParentRight="true"
    android:layout_alignParentEnd="true"
    android:padding="16dp"
    android:minHeight="48dp" />
<!-- EditText for entering password -->
<EditText android:id="@+id/editText2"
    android:layout_width="match_parent"
    android:layout_height="48dp"
    android:inputType="textPassword"
    android:ems="10"
    android:layout_below="@+id/editText"
    android:layout_alignParentLeft="true"
    android:layout_alignParentStart="true"
    android:layout_alignRight="@+id/editText"
    android:layout_alignEnd="@+id/editText"
    android:hint="Password"
    android:padding="16dp"
    android:minHeight="48dp" />
<TextView android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/editText2"
    android:layout_alignParentStart="true"
    android:layout_alignParentLeft="true"
    android:text="Attempts Remaining"
    android:textSize="25dp" />
<TextView android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignTop="@+id/textView2"
    android:layout_alignBottom="@+id/textView2"
    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    android:layout_toEndOf="@+id/textview"
    android:layout_toRightOf="@+id/textview"
    android:text="Text"
    android:textSize="25dp" />
<Button android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="login"
        android:id="@+id/button"
        android:layout_alignParentBottom="true"
        android:layout_toLeftOf="@+id/textview"
        android:layout_toStartOf="@+id/textview" />
    <Button android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Cancel"
        android:id="@+id/button2"
        android:layout_alignParentBottom="true"
        android:layout_toRightOf="@+id/textview"
        android:layout_toEndOf="@+id/textview" />
</RelativeLayout>
```

Practical:-11

A] Create an android application using linear layout and insert COUNTRY NAMES in the list view and display the selected game in the text view.

In Android, a LinearLayout is a view group that arranges its child elements either vertically or horizontally. It is used to create simple layouts where each child view is placed in a single row or column. You can set the orientation using android:orientation="vertical" or "horizontal".

- Create a New Project
- Open activity_main.xml
- Replace ConstraintLayout with LinearLayout
- Add Child Views
- Set Layout Properties

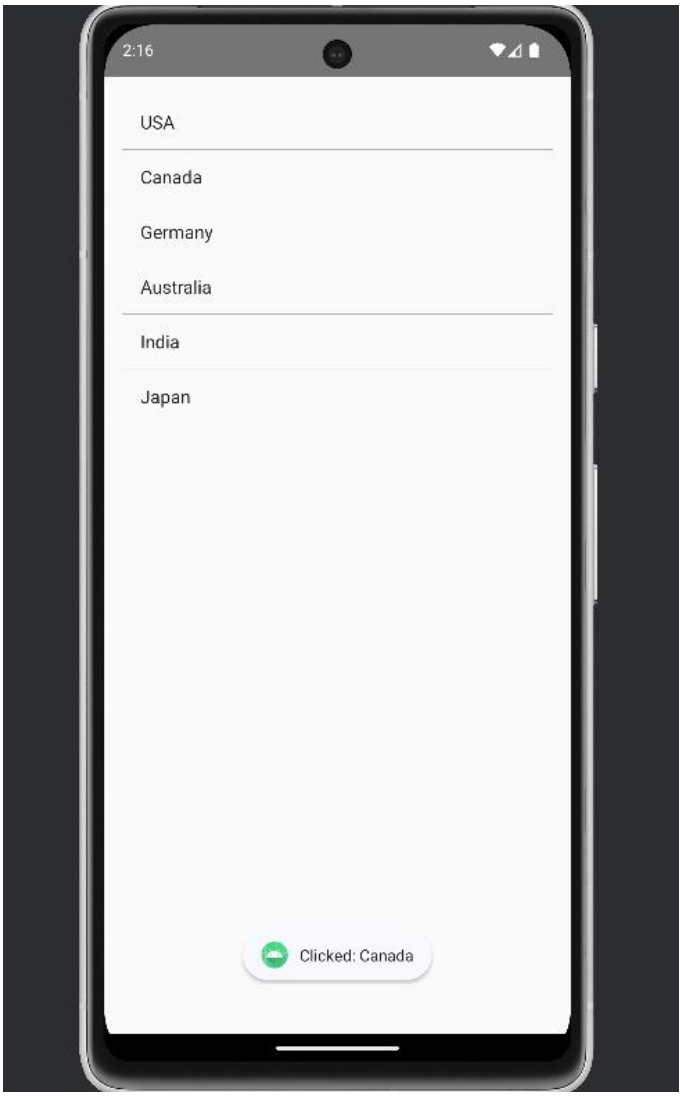
(MainActivity.java)

```
package com.example.listviewexample;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   // Sample data for the list
   String[] countries = {"USA", "Canada", "Germany", "Australia", "India", "Japan"};
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      // Get a reference to the ListView
      ListView listView = findViewById(R.id.listView);
      // Create an ArrayAdapter to bind the data to the ListView
      ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1, // Use a built-in layout for list items
            countries
      );
      // Set the adapter on the ListView
      listView.setAdapter(adapter);
      // Set an item click listener to handle item clicks
      listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
         @Override
         public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
            // Show a Toast message with the clicked item
            String clickedItem = (String) parent.getItemAtPosition(position);
            Toast.makeText(MainActivity.this, "Clicked: " + clickedItem,
Toast.LENGTH_SHORT).show();
```

```
        }
    });
  }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent">
  <ListView
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:divider="@android:color/darker_gray"
    android:dividerHeight="0.5dp"
    android:padding="16dp"/>
</RelativeLayout>
```



Practical:-12

Q] Create an android application to demonstrate the following event list eners on the following widgets Button: on click listener, long click listener Image: Touch listener with these motion events =>Action up, action down and action pointer down Edit Text : key listener on Enter key The Appropriate message be displayed in text view

Event listeners in Android are used to detect and respond to user interactions like clicks, touches, or key presses. They are implemented by defining a method in the listener interface (e.g., OnClickListener, OnTouchListener) and associating it with a UI element, enabling dynamic behavior in the app.

- Create a New Android Project
- Open activity_main.xml
- Implement Event Listener in MainActivity.java or MainActivity.kt
- Choose the Right Event Listener
- Run the program

(MainActivity.java)

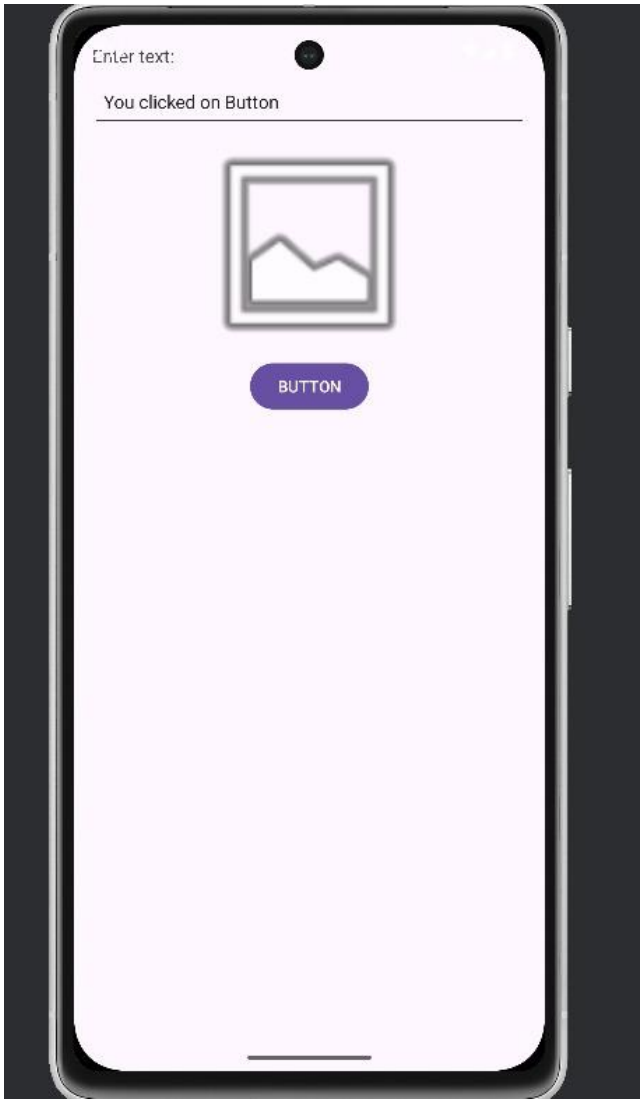
```
package com.example.slip2;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.os.Bundle;
import android.view.KeyEvent;
import android.view.MotionEvent;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    private TextView textView;
    private Button button;
    private ImageView imageView;
    private EditText editText;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        textView = findViewById(R.id.editText);
        button = findViewById(R.id.button);
        imageView = findViewById(R.id.imageView);
        editText = findViewById(R.id.editText);
        // Button Click Listener
        button.setOnClickListener(v -> textView.setText("You clicked on Button"));
```

```
        // Button Long Click Listener
        button.setOnLongClickListener(v -> {
            textView.setText("You long clicked on Button");
            return true;
        });
        // Image Touch Listener
        imageView.setOnTouchListener((v, event) -> {
            switch (event.getActionMasked()) {
                case MotionEvent.ACTION_DOWN:
                    textView.setText("ACTION DOWN");
                    break;
                case MotionEvent.ACTION_UP:
                    textView.setText("ACTION UP");
                    break;
                case MotionEvent.ACTION_POINTER_DOWN:
                    textView.setText("ACTION POINTER DOWN");
                    break;
            }
            return true;
        });
        // EditText Key Listener (Enter Key)
        editText.setOnKeyListener((v, keyCode, event) -> {
            if (keyCode == KeyEvent.KEYCODE_ENTER && event.getAction() ==
KeyEvent.ACTION_DOWN) {
                textView.setText("Enter key pressed");
                return true;
            }
            return false;
        });
    }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:id="@+id/editTextLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Enter text:"
        android:textSize="16sp"
        android:layout_marginBottom="4dp"/>
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
```

```
        android:layout_height="50dp"
        android:padding="10dp"
        android:textSize="16sp"
        android:hint="Type and press Enter"
        android:labelFor="@id/editTextLabel"
        android:importantForAccessibility="yes" />
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@android:drawable/ic_menu_gallery"
        android:layout_gravity="center" />
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BUTTON"
        android:layout_gravity="center" />
</LinearLayout>
```



Practical:-13

Q] Create an android application that opens the website in the browser on the click of a button using intents.

To create an Android app that opens a website in a browser when a button is clicked, you use an Intent. Set the button's OnClickListener to trigger an Implicit Intent with the ACTION_VIEW action and the website URL, which opens the URL in the default web browser.

- Create a New Android Project
- Open activity_main.xml
- Set Up Button Event Listener in MainActivity.java or MainActivity.kt
- Build and Run the Application

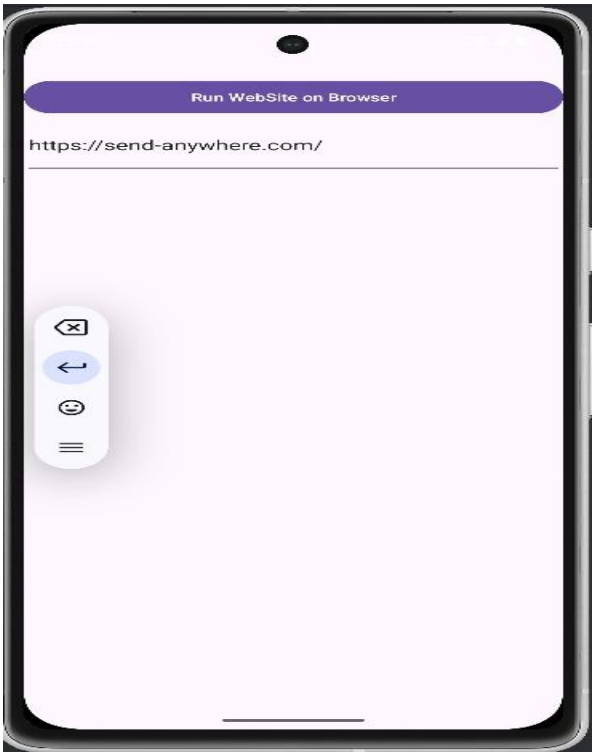
(MainActivity.java)

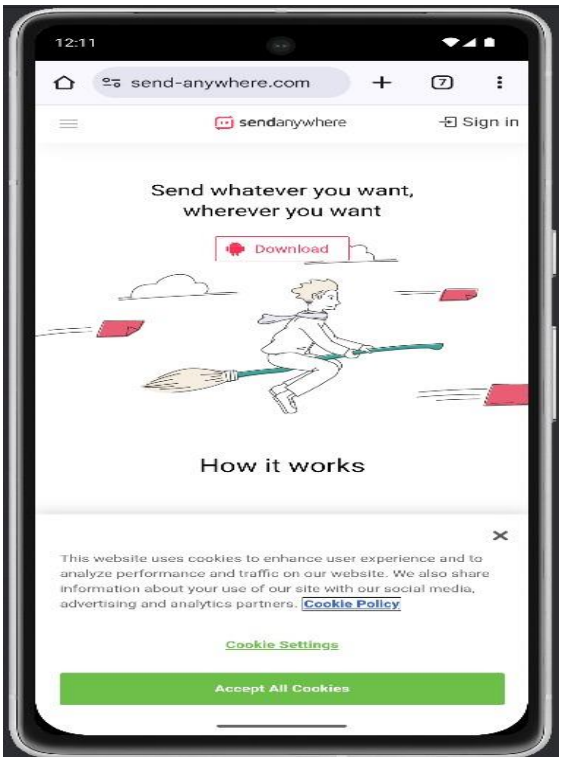
```
package com.example.openurl;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.graphics.Color;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity {
   @Override
protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       Button runWebsite = (Button)
             findViewById(R.id.runWebsite);
       runWebsite.setOnClickListener(new View.OnClickListener() {
          public void onClick(View view) {
             EditText website = (EditText)
                   findViewById(R.id.website);
             String strURL =
                  website.getText().toString();
             if
             (strURL.indexOf("http://www") < 0) {
```

```
                strURL = "http://www" + strURL;
            }
            Intent implicit = new Intent(Intent.ACTION_VIEW, Uri.parse(strURL));
            startActivity(implicit);
        }
    });
}}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
   tools:context=".MainActivity">
   <LinearLayout android:layout_width="match_parent" android:layout_height="match_parent"
android:orientation="vertical">
      <TextView
         android:id="@+id/openURL"
         android:layout_width="match_parent"
         android:layout_height="70dp"
         android:rotationY="40"
         android:text="OPENURL" />
      <Button android:id="@+id/runWebsite" android:layout_width="fill_parent"
android:layout_height="50dp"
         android:text="Run WebSite on Browser" />
      <EditText
         android:hint="Type URL Here" android:id="@+id/website" android:layout_width="fill_parent"
android:layout_height="75dp" android:ems="5"></EditText>
   </LinearLayout>
</RelativeLayout>
```





Practical:-14

Q] Create an android application to generate any random number with some time delay using thread.

To create an Android app that generates a random number with a time delay, you can use a Thread to introduce the delay. The thread will run a background task, generate a random number, and then update the UI with the number after the delay using Handler.post() or runOnUiThread().

- Create a New Android Project
- Design the Layout in activity_main.xml
- Write Code for Random Number Generation with Time Delay
- Run the App

(MainActivity.java)

```
package com.example.random_no_generator;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
 import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.QuickContactBadge;
import android.widget.TextView; import android.widget.Toast; import java.util.Random;
public class MainActivity extends AppCompatActivity { Button btn1;
   TextView tv1; int i=0;
   int ran=0; @Override
   protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
      btn1 = (Button) findViewById(R.id.btnRandom);
      tv1 = (TextView) findViewById(R.id.lbl1);
      btn1.setOnClickListener(new View.OnClickListener() { @Override
      public void onClick(View v) {
         try
         {
            int ran = (int) (Math.random()*100);
            tv1.setText("Number=" + ran);
         }
         catch(Exception ex)
         {
         }
      }
      });
```

```
    }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
android:layout_height="match_parent"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">
        <TextView
            android:id="@+id/lbl1"
            android:layout_width="match_parent"
            android:layout_height="55dp"
            android:text="Output"
            android:textColor="@color/red"
            android:textSize="24sp"
            android:textStyle="bold" />
        <Button
            android:id="@+id/btnRandom"
            android:layout_width="match_parent"
            android:layout_height="58dp"
            android:background="@color/yellow"
            android:text="Generate Random" />
    </LinearLayout>
</RelativeLayout>
```